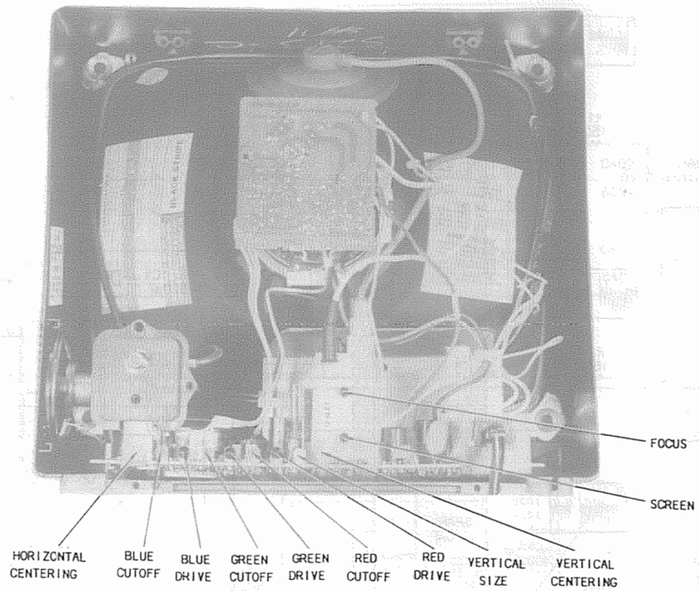


CABINET–REAR VIEW

DISASSEMBLY INSTRUCTIONS

CHASSIS REMOVAL

Remove six screws holding cabinet back and remove back. Disconnect HV anode, CRT socket, Deflection yoke connectors, Degaussing coil connectors, Speaker connectors, Ground leads, and all other required cabling. Release two latches holding main board assembly to cabinet bottom and slide board assembly from cabinet.

CRT REMOVAL

(Caution: Some sets employ a CRT with neck assemblies permanently bonded to CRT. DO NOT attempt to remove these assemblies.)

Follow chassis removal procedure and lay set face down on a soft protective surface. Loosen and remove CRT neck assemblies. (See Caution). Remove four screws holding CRT to cabinet front and lift CRT out of cabinet. DO NOT LIFT CRT BY NECK.

SERVICING IN THE FIELD

CRT IMPLOSION PROTECTION AND CLEANING

Implosion protection is an integral part of the picture tube, cleaning accomplished without CRT removal.

FUSE DEVICES

A 4-amp fuse is used for AC line protection. (See photo, Main Board.)

VHF/UHF TUNER

See Miscellaneous Adjustments.

CHANNEL TUNING

Channel Up and Down buttons are provided for channel scanning. Fine tuning is automatic.

HORIZONTAL OSCILLATOR

Adjustment of the horizontal hold is accomplished by the proper setting of the Horizontal Frequency Control.

HIGH VOLTAGE

For high voltage procedure, refer to Miscellaneous Adjustments.

FOCUS

The focus may be varied by a Focus Control. (See photo, Cabinet - Rear View.)

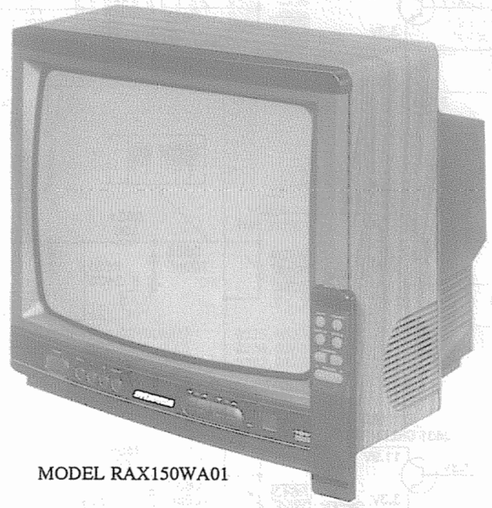
AGC

The RF AGC may be varied by an RF AGC Control. (See photo, Main Board.)

CENTERING

Horizontal centering is accomplished by proper adjustment of the horizontal centering control. (See photo, Cabinet - Rear View.)

Vertical centering is accomplished by proper adjustment of the vertical centering control. (See photo, Cabinet - Rear View.)



MODEL RAX150WA01

MODEL	CHASSIS
RAX150WA01	13L404
RAX150WA02	13L404
RAX156CH01	13L404
RAX156CH02	13L404
RKX185BK01	20L101

SAFETY PRECAUTIONS

See Page 1

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SAMS™ Howard W. Sams & Company
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The listing of any available replacement part herein does not constitute in any case a recommendation, warranty or guaranty by Howard W. Sams & Co. as to the quality and suitability of such replacement part. The numbers of these parts have been compiled from information furnished to Howard W. Sams & Co. by the manufacturers of the particular type of replacement part listed.

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91RG02099 Printed in U.S. of America
DATE 5-91 SET 2827 FOLDER 2

SYLVANIA
CHASSIS 13L404, 20L101
SET 2827 FOLDER 2

SAFETY PRECAUTIONS

SERVICE WARNING

Service work should be performed only by qualified service technicians who are familiar with safety checks and guide lines.

1. For continued safety, no modification of any circuit should be attempted unless recommended by manufacturer.

2. Disconnect power source before replacing parts as some parts may be electrostatic sensitive.

3. Use an isolation transformer between the line cord and power receptacle, when servicing chassis.

SERVICING HIGH VOLTAGE AND PICTURE TUBE

When servicing the High Voltage circuits, extreme caution should be used.

1. Discharge static High Voltage by connecting a 10 kohms resistor in series with a test lead between chassis and anode lead of picture tube.

2. Wear shatter-proof eye protection (goggles) when handling the picture tube in case of implosion.

3. DO NOT lift picture tube by the neck.

X-RAY RADIATION AND HIGH VOLTAGE LIMITS

Service personnel should be aware of the procedures and instructions covering x-ray radiation. The only potential source of x-ray in present day solid state receivers and monitors is the picture tube.

1. It is only when High Voltage is excessive that x-ray radiation is capable of being emitted from shell of picture tube. Be sure the High Voltage is set at specified level.

2. An accurate High Voltage meter should be available at all times. Meter calibration should be checked periodically.

3. High Voltage should be kept at rated value - NO HIGHER. Higher voltages may cause x-ray radiation or failure of other associated components. DO NOT depend on protection circuit to keep voltages at rated value.

4. Every time a chassis is serviced, High Voltage should be checked at various brightness levels to be sure it is regulating properly.

5. While troubleshooting a set with excessive High Voltage, avoid being close to picture tube. DO NOT operate longer than it is necessary to locate the cause of excessive High Voltage. Use a variable AC transformer to regulate voltage.

6. Many components, electrical and mechanical, in present chassis have safety related characteristics which are not evident with visual inspection. When these components are known, they are identified with a # on the schematic and in the parts list. When replacing these components, for SAFETY, use only an equivalent replacement part.

SAFETY CHECKS-FIRE AND SHOCK HAZARD

Cold Leakage Checks (Sets with isolated ground.)

1. Unplug the AC cord and connect a jumper across the two prongs on the plug.

2. Turn on power switch.

3. Measure the resistance, with an Ohm meter, between the jumpered AC plug and any exposed metal cabinet parts on the set such as: antenna screw heads, control shafts, handle brackets. Exposed metal parts that have a return path should measure between 200 kohms and 5 megohm. Parts without a return path must measure infinity.

Leakage Current Hot Check

1. Plug the AC cord directly into AC outlet. DO NOT use an isolation transformer.

2. Connect a 1500 Ohm 10 watt resistor, in parallel with a .15μF 150V AC capacitor, between any exposed metal parts on the set and a good earth ground such as a water pipe. (See Figure below.)

3. Using an AC volt meter, with 1000 Ohms per volt or more sensitivity, measure the voltage across the resistor. Check each exposed part and measure voltage at each point.

4. Reverse the AC plug and repeat voltage measurement at each point.

5. The voltage at any point should not exceed .75 volts RMS. This corresponds to .5 milliamps AC. Any value exceeding this limit constitutes a potential shock hazard and must be corrected.

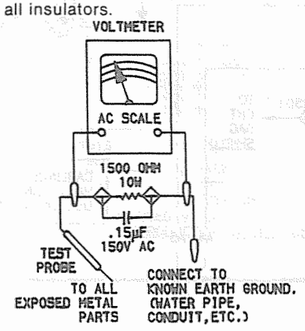
GENERAL GUIDE LINES

A final SAFETY check before returning the set to customer.

1. Check area repaired for poorly soldered or de-soldered connections. Check entire circuit board surface for solder splashes.

2. Check interboard wiring for pinched wires or wires contacting any high-wattage resistors.

3. Check that all control knobs, shields, covers, grounds and mounting hardware have been replaced. Be sure to replace all insulators.



TROUBLESHOOTING AID

Note: Waveforms taken with triggered scope, Keyed-Rainbow generator. Schematic voltages measured with digital meter, no signal. Controls adjusted for normal operation.

PICTURE or SOUND

NO PIC, NO SOUND, NO RASTER: Check AC power supply and sources generated from Horizontal Output Transformer (T501). Refer to "Troubleshooting" Power Supply and Horizontal circuits.

NO PIC, NO SOUND, HAS RASTER: Check IF-AGC and source voltages from Horizontal Output Transformer (T501). Refer to "Troubleshooting" IF-AGC and Horizontal circuits.

NO PIC, HAS SOUND, NO RASTER: Check Horizontal Output Transformer (T501) sources and Video circuit. Refer to "Troubleshooting" Horizontal and Video circuits.

NO PIC, HAS SOUND, HAS RASTER: Refer to "Troubleshooting" Video circuit.

HAS PIC, NO SOUND: Refer to "Troubleshooting" Audio circuit.

OVERLOADED PICTURE: Refer to "Troubleshooting" IF-AGC circuit.

LOW OR EXCESSIVE BRIGHTNESS: Check Video and Luminance circuits. Refer to "Troubleshooting" Video circuit.

SWEEP

NO RASTER, HAS SOUND: Check HV rectifier, Part of Horizontal Output Transformer (T501). Refer to "Troubleshooting" Horizontal circuit.

NO RASTER, NO SOUND: Refer to "Troubleshooting" Horizontal circuit.

NO VERT DEFLECTION: Refer to "Troubleshooting" Vertical circuit.

POOR VERT LIN OR FOLDOVER: Refer to "Troubleshooting" Vertical circuit.

POOR HORIZ LIN OR FOLDOVER: Refer to "Troubleshooting" Horizontal circuit.

NARROW PICTURE: Refer to "Troubleshooting" Horizontal circuit.

VERT OFF FREQUENCY: Refer to "Troubleshooting "Vertical circuit.

HORIZ OFF FREQUENCY: Refer to "Troubleshooting" Horizontal circuit.

SYNC

NO VERT/HORIZ SYNC: Refer to "Troubleshooting" Sync circuit.

RASTER

YELLOW (NO BLUE): Check Chroma and Blue Output circuits. Refer to "Troubleshooting" Raster circuit.

CYAN (NO RED): Check Chroma and Red Output circuits. Refer to "Troubleshooting" Raster circuit.

MAGENTA (NO GREEN): Check Chroma and Green Output circuits. Refer to "Troubleshooting" Raster circuit.

COLOR (B/W operating normally)

NO COLOR: Refer to "Troubleshooting" Chroma circuit.

WEAK COLOR: Refer to "Troubleshooting" Chroma circuit.

NO COLOR SYNC: Refer to "Troubleshooting" Chroma circuit.

NO GREEN: Check Chroma and Green Output circuits. Refer to "Troubleshooting" Raster circuit.

NO BLUE: Check Chroma and Blue Output circuits. Refer to "Troubleshooting" Raster circuit.

NO RED: Check Chroma and Red Output circuits. Refer to "Troubleshooting" Raster circuit.

INCORRECT HUE (TINT): Refer to "Troubleshooting" Chroma circuit.

TROUBLESHOOTING

POWER SUPPLY

Check the AC Fuse (F400) and DC Fuse (F401). If Fuse F400 is open, check Bridge Rectifier Diodes (D401 thru D404), Capacitors C400 thru C404, C412 and Electrolytic C405. If Fuse F401 is open, check Power Regulator IC (IC430) and Horizontal Output Transistor (Q502). Apply 120V and check for 150 V at the cathode of D402. If 150V is missing at the cathode of D402, check Line Filter (L400) and Power Relay (K400) and Resistors R402, R403. If 150V is present at the cathode of D402, check for 130V at TP4. If this voltage is missing, check voltages and components associated with IC430, Resistors R438, R439 and Q502. If the proper voltage is present at TP4, refer to the "Horizontal" section of this Troubleshooting guide.

AUDIO

Select an active TV channel and check for an audio waveform at pin 12 of the TV Signal Processor IC (IC200. If there is no audio, check the voltages, waveforms and components associated with pins 11 thru 15 of IC200. If audio is present, check for audio at pin 2 of the Audio Output IC (IC230). If audio is missing check the voltages, waveforms and components associated with IC230. Check the voltage at pin 11 of IC200, it should measure 1.1V at mute and 2.6V at Maximum volume.

VIDEO

Inject a video signal at TP1 and check for video on the CRT. If video is present, refer to the "IF-AGC" section of this Troubleshooting guide. If there is no video on the CRT, check for a video waveform at pin 10 of the Chroma/Lum IC (IC600). If video is missing at pin 10, check the voltages, waveforms and components associated with EQUL Transistor (Q602) and the Peaking Buffer Transistor (Q603). If video is present at pin 10, check for a video waveform at pins 12, 13, 14 of IC600. If the waveform is missing, check the voltages, waveforms and components associated with pins 1 thru 4, 6, 9, 10 thru 16 of IC600. If the waveform is present at pins 12, 13 and 14, check the voltages, waveforms and components associated with the Output Transistors (Q1, Q2, Q3). If the brightness is inadequate or cannot be controlled, check the voltages, waveforms and components associated with pin 9 of IC600 and pin 8 of the CRT.

IF-AGC

Inject a video IF signal at the IF input and check for video on the CRT. If video is present, check the tuner, tuner control and tuner AFC circuits. If there is no video on the CRT, check for a video waveform at TP12. If video is present at TP12, refer to the "Video" section of this Troubleshooting guide. If there is no video at TP12, apply AGC bias to pin 10 of the TV Signal processor IC (IC200). If video is now present at TP12, check the voltages, waveforms and components associated with pins 1, 5 and 10 of IC200. If there is still no video at TP12, check the voltages, waveforms and components associated with pins 1, 5, 6, 8, 9, 10, 16 thru 21 of IC200. A defective AGC circuit can cause an overloaded picture, excessive snow or loss of audio and video. See the AGC Voltage Chart for AGC voltages with signal.

AGC VOLTAGE CHART
IC200

Pin 1	6.4V
Pin 5	6.3V
Pin 10	2.4V

CHROMA

Check for a chroma waveform at pin 3 of the Chroma/Lum IC (IC600). If the waveform is missing, check the components associated with pin 3. If a chroma waveform is present at pin 3 of IC600 , check for the proper waveforms at pins 12, 13, 14 of IC600. If these waveforms are missing, check the voltages, waveforms and components associated with pins 5, 7, 8, 12, 13, 14, 17 thru 20 of IC600. Check the 3.58MHz oscillator at pin 18 of IC600. Check the voltages and components associated with the color control and pin 5 of IC600. If there is inadequate tint range, check the voltages, waveforms and components associated with the tint control and pin 17 of IC600. If the proper waveforms are present at pins 12, 13, 14 of IC600, refer to the "Raster" section of this Troubleshooting guide.

HORIZONTAL

Determine if the TV is in shutdown, refer to the "High Voltage Shutdown" section, of this Troubleshooting guide. If the TV is not in shutdown, inject a horizontal signal at the base of the Horizontal Output Transistor (Q502). If horizontal deflection is now present, check the voltages, waveforms and components associated with pins 23, 24, 26, 27, 28 of the TV Signal Processor IC (IC200) and the Horizontal Driver Transistor (Q501). If there is still no horizontal sweep, check the voltages, waveforms and components associated with Q502 and the Horizontal Output Transformer (T501). Check the voltages and components associated with Diodes D409, D445 and D460 for defects. The high voltage rectifier is part of Transformer T502 and if defective will affect the performance of the horizontal circuits. If the Horizontal Oscillator is off frequency, check the voltages, waveforms and components associated with pins 23 and 24 . Horizontal linearity or foldover problems may be caused by Capacitors C505, C506 and C507 being defective.

HIGH VOLTAGE SHUTDOWN

The high voltage is monitored by Diode D530, rectifying pulses from the Horizontal Output Transformer (T501). Should the high voltage increase, the rectified voltage at the cathode of Diode D530 will also increase and trigger the Over Voltage Shutdown Latch Transistors (Q530, Q531). To troubleshoot, remove Diode D530 from the circuit. Use a variac for AC power and start at 90VAC and increase as necessary to locate and repair that defect. Return D530 to the circuit.

TROUBLESHOOTING (Continued)

NOTE: Care should be taken in defeating the high voltage shutdown circuit, as this may cause excessive X-radiation and damage to the CRT, Transformer T501 and associated components. Monitor the high voltage and troubleshoot.

Voltages Taken with TV in Shutdown
IC430

Pin 4	0V

HIGH VOLTAGE SHUTDOWN TEST

Apply 120V AC, turn set on, set all customer controls for normal operation and apply a short across Zener Diode Z530. Set should lose raster and sound. If set does not lose raster and sound the shutdown circuit should be repaired. To resume normal operation, remove AC Power and wait 30 seconds then turn set On.

VERTICAL

Inject a vertical drive signal at pin 3 of the TV Signal Processor IC (IC200). If vertical deflection is now present, check voltages, waveforms and components associated with pins 2, 3, 4 of IC200. If there is still no vertical deflection, check the voltages, waveforms and components associated with the Vertical Output IC (IC550). Vertical linearity or foldover problems maybe caused by vertical feedback and bias circuits, check electrolytics C551, C553, C554, C558 for defects.

SYNC

If there is no vertical or horizontal sync, check the voltages, waveforms and components associated with pins 22, 25 of the TV Signal Processor IC (IC200). If there is no vertical sync, check the voltages, waveforms and components associated with pins 2, 3, 4 and 25 of IC200. If there is no horizontal sync, check the voltages and components associated with pins 23, 24 and 25 of IC200.

RASTER

Check the CRT and CRT voltages. If there is no Red, check the voltages and components associated with pin 12 of the Chroma/Lum IC (IC600) and Red Output Transistor (Q1). If there is no Green, check the voltages and components associated with pin 13 of IC600 and Green Output Transistor (Q2). If there is no Blue, check the voltages and components associated with pin 14 of IC600 and Blue output Transistor (Q3). If the raster has a keystone shape, check the Deflection Yoke (L500). If the raster has height or width problems, refer to the "Vertical", "Horizontal" and "Power" sections of this Troubleshooting guide.

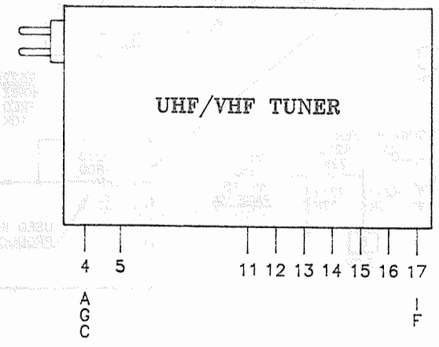
TUNER VOLTAGE CHART

AGC

	4	5	11	12	13	14	15
VHF Low Band	9.2V	12.0V	1.0V	4.9V	.1V	5.0V	0V
VHF High Band	9.2V	12.0V	2.3V	4.9V	.1V	5.0V	0V
UHF Band	9.2V	12.0V	1.6V	4.9V	.1V	5.0V	0V

NOTE: VHF Low Band voltages taken on channel 2.
VHF High Band voltages taken on channel 7.
UHF Band voltages taken on channel 14.

TUNER TERMINAL GUIDE



TEST EQUIPMENT

Test Equipment listed by Manufacturer illustrates typical or equivalent equipment used by SAMS' Engineers to obtain measurements and is compatible with most types used by field service technicians.

Equipment	B&K Precision Equipment No.	Sencore Equipment No.	Notes
OSCILLOSCOPE	1541A, 2120, 2125, 2160	SC61	
GENERATORS			
RGB	1249, 1260	RG67	
MULTIBURST SIGNAL	1251, 1260	VA62A	
COLOR BAR	1211A, 1249, 1251, 1260	VA62A, CG25, NT64	
ANALOG VOM	114, 117, 177, 214		
DIGITAL VOM	388HD, 2900 SERIES	DVM37, DVM56A, SC61	
FREQUENCY METER	1803, 1804, 1805	FC71, SC61	
HI-VOLTAGE PROBE VOM/DMM Accessory probes	HV-44 PR-28(HV)	HP200 TP212	
ISOLATION TRANSFORMER	TR110, 1604, 1653, 1655	PR57	
CAPACITANCE ANALYZER	820, 810, 830	LC76, LC101, LC102	
CRT ANALYZER	467, 470, 480, 490	CR70	
TEMPERATURE PROBE	TP-28, TP-30		
AC LEAKAGE TESTER	1655	PR57	
LOGIC PROBE	DP51, DP21		
LOGIC PULSER	DP101, DP31		
INDUCTANCE ANALYZER	875A	LC76, LC101, LC102	
FLYBACK YOKE TESTER	875A	VA62A, LC76, LC101, LC102	
TV STEREO GENERATOR	2009	ST65, ST66	
TV STEREO POWER MONITOR		SR68	
FIELD STRENGTH METER		FS73, FS74	
TRANSISTOR TESTER		TF46	
VIDEO ANALYZER		VA62A	

TV ALIGNMENT INSTRUCTIONS

Use an isolation transformer and observe power supply polarity. Maintain line voltage at 120V AC. Allow a 20-minute warm-up period for receiver and test equipment.
Suggested Alignment Tools: GC-THORSEN

None

PRELIMINARY INSTRUCTIONS

Set the channel selector to the highest unused channel. Set scope sweep to external. Connect scope vertical input to scope vertical input on sweep/marker generator. Connect scope external horizontal input to scope horizontal input on sweep/marker generator. Ground test equipment to TV chassis unless specified otherwise. Use only enough generator output to provide a usable indication.
Note: Response may vary slightly from that shown.

Connect 2.5V Bias to TP210.

VIDEO IF ALIGNMENT (SWEEP MARKER GENERATOR)

DIRECT PROBE FROM SWEEP/MARKER GENERATOR	SWEEP GENERATOR OUTPUT	SWEEP GENERATOR FREQUENCY	MARKER GENERATOR FREQUENCY	REMARKS
TP1	TP209	44MHz (10MHz Sweep)	45.75MHz	Adjust L202 to place 45.75 MHz marker at Maximum. See Figure 1.

TV ALIGNMENT INSTRUCTIONS (Continued)

VIDEO IF ALIGNMENT (BAR SWEEP GENERATOR)

BAR SWEEP GENERATOR	SCOPE INPUT	REMARKS
Antenna Terminal or TP209	TP1	Perform Video IF Adjustments per SWEEP/MARKER GENERATOR instructions. See Figure 2.

SOUND IF ALIGNMENT

Tune in a station and adjust L221 for maximum sound. Reduce signal strength at the antenna terminals until distortion appears. Continue to reduce the signal while aligning for undistorted output by adjusting L221.

AUTOMATIC FINE TUNING ALIGNMENT

Connect as explained in preliminary instructions unless specified otherwise.

DIRECT PROBE FROM SWEEP/MARKER GENERATOR	SWEEP GENERATOR OUTPUT	SWEEP GENERATOR FREQUENCY	MARKER GENERATOR FREQUENCY	REMARKS
TP1	TP2	44MHz (10MHz Sweep)	45.75MHz	Adjust L210 to place 45.75MHz marker at Maximum. See Figure 3.

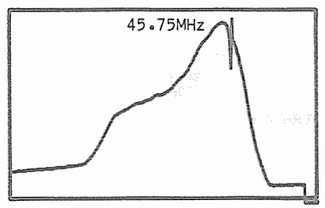


Figure 1

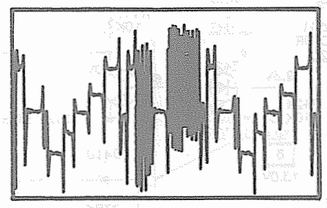


Figure 2

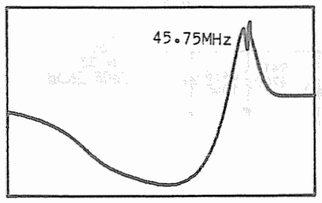


Figure 3

A

B

SEE IC350 PIN 9 PAGE 3A

TP2

C367 .01

R363 100K

R364 100

C364 56pF

C363 .27

L210

ANTENNA ISOLATOR

P001

TO RF INPUT UHF/VHF TUNER

R362 82K

R365 82K

8 13.0V

C213 56pF NPO

C215 .022

TP209

FROM 17 UHF/VHF TUNER

Y200

FILTER

PART OF TDA8305 IC200 TV SIGNAL PROCESSOR

L203

R201 22

C202 4.7pF NPO

A001 VERSIONS ONLY

C203 10pF NPO

L202

TUNERS NOT INCLUDED IN THIS COVERAGE.

TO 4 UHF/VHF TUNER

RF AGC RANGE 4.3V TO 9.32V

C208 22µF

R209 27K

R208 10K

R206 RF AGC 22K

R205 3300

R207 4700

C206 .022

C214 1µF

R214 150K

TP210

IF AGC RANGE 2.1V TO 2.4V

TO 15 UHF/VHF TUNER

R369 10 (47K)

TO 12 UHF/VHF TUNER

L365 1.5

C365 100µF

9 5.0V

TO 5 UHF/VHF TUNER

R360 15K

C360 100µF

TO 16 UHF/VHF TUNER

TO 11 UHF/VHF TUNER

R366 22K

C362 .001

C361 22µF

Z361 33.0V 2W

R361 15K

1 130V

Q210 COIN BUFFER

C212 56pF NPO

C210 .1

C211 56pF NPO

R211 15K

R210 22K

R212 27K

SEE IC350 PIN 16 PAGE 3A

C602 .047

R603 1800

C603 150pF

PART OF TDA8305 IC200 TV SIGNAL PROCESSOR

C465 470µF

R465 6.2

R514 18K

R520 1000

C520 .01

C521 10µF

R523 82K

R521 27K

C522 .0033

C522 2.7µF

R522 HORIZ FREQ 10K

C518 .015

R524 HORIZ CENTERING 47K

R525 18K

R526 82K

C523 .22

R513 1800

SEE Q327 BASE PAGE 3A

USED IN EARLY PRODUCTION MODELS ONLY

A1

A

Y220 4.5MHz

FILTER

C219 .022

C216 22pF NPO

C218 47pF NPO

PART OF TDA8305 2.3V IC200 TV SIGNAL PROCESSOR 2.3V

C221 .022

C222 .022

L221

C241 47µF

SEE Q330 EMITTER PAGE 3B

SEE Q421 EMITTER PAGE 2E

L220

C220 27pF NPO

L601

R601 82

C601 150pF

L602

Y601 4.5MHz

FILTER

TP1

Q612 VIDEO BUFFER

R602 390

R612 680

8 13.0V

TAKEN WITH BAR SWEEP GENERATOR

C609 27pF NPO

C614 47pF NPO

R614 2000

L614 1.5

C613 .01

R601 82

TDA3569 IC600 CHROMA/LUM

C619 .01

C610 220µF (470µF)

R609 10

A001 VERSION ONLY

C612 2.2µF

C615 .1

R623 10K

C622 .0056

C623 .22

C628 560pF

R230 7500 2%

C230 .047

R233 2000 2%

Q240 MUTING

C232 .33µF

C234 .001

TDA2611A AUDIO OUTPUT

C235 22µF

C231 1000µF

R231 15

6 16.0V

NC NC NC

C237 220µF

P/J200

C236 .1

R232 3.3

SEE IC350 PIN 19 PAGE 3A,3B

R240 100K

C240 .047

C550 390pF

LA7830 IC550 VERT OUTPUT

R550 3900

C555 .0068

R551 330

C556 .01

D550

C551 100µF

SEE Q338 BASE PAGE 3A

C552 .033

R553 390

C553 100µF (680µF)

R554 20K

R556 10K

R558 15K

C558 10µF

R559 VERT SIZE 200

C554 4.7µF

R557 510

R555 5600

R560 2.4

R561 3.6

R562 1200

R545 VERT CENTERING 10K

C565 .22

R565 390K

L503

R501 150

Q501 HORIZ DRIVER

C501 100pF

R502 47K

C509 220pF

C502 470pF

C503 560pF

R503 2400

R504 2700 (6800) 3W

C504 .1

A001 VERSIONS ONLY

B002 VERSIONS ONLY

T502

R505 .68 1W

L501

USED IN EARLY PRODUCTION MODELS ONLY

2SD1554 Q502 HORIZ OUTPUT

C505 3.9pF

C506 .001

R511 300K

R343 10K J TEST

NC

C507 .36 (.27)

DEFLECTION YOKE L500

14.3

5.4

P/J11

P/J10

PJ11

B506

L507

MAGNET R507 680

R532 1M

R533 2740 1%

R534 1210 1%

R538 1M

R530 15K (6200)

R531 10K

D530

C530 10µF

A1 A2 A3

A PHOTOFACT STANDARD NOTATION SCHEMATIC WITH CIRCUITRACE

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C

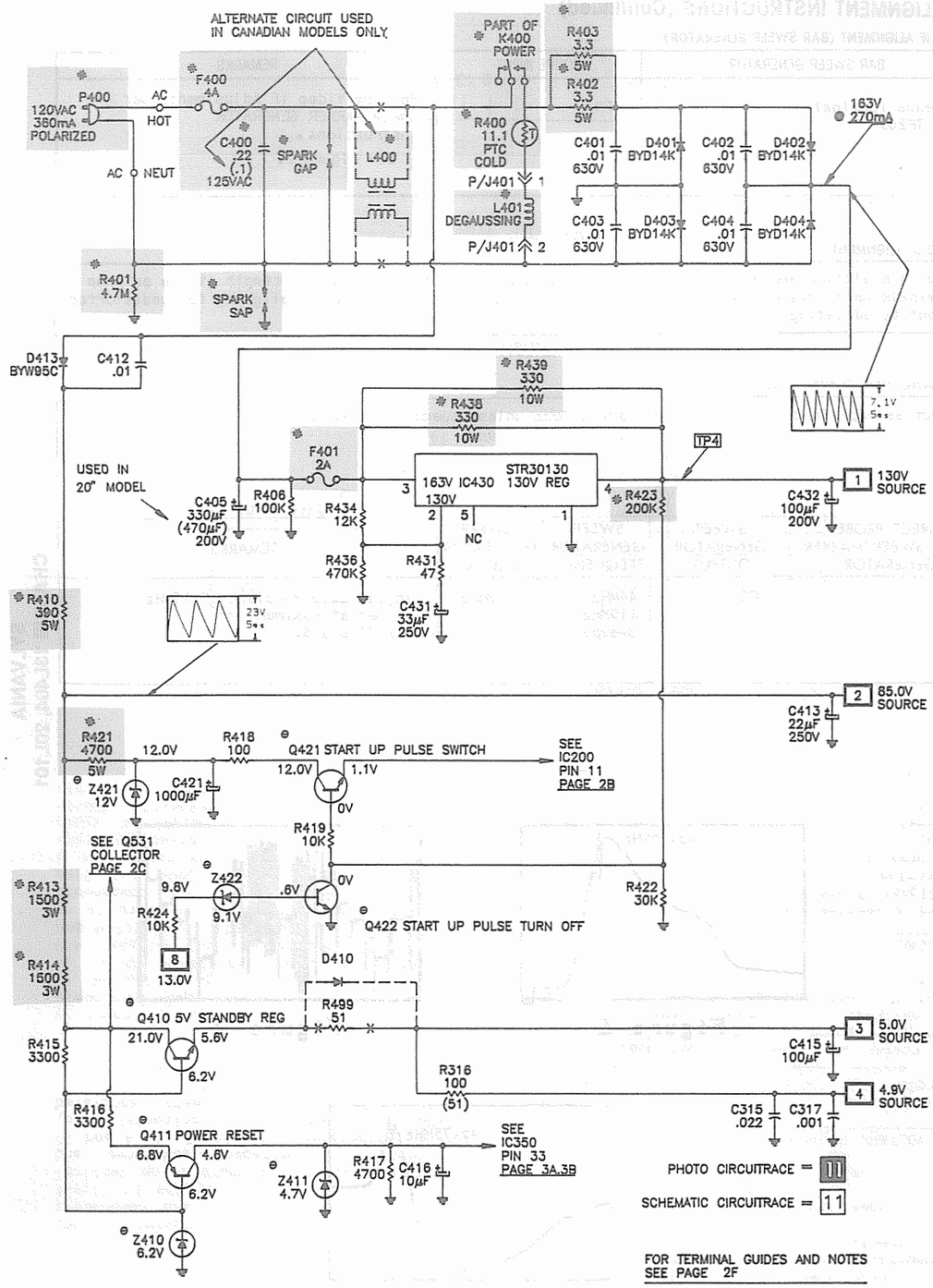
D

PHOTO CIRCUITRACE = 11

SCHEMATIC CIRCUITRACE = 11

FOR TERMINAL GUIDES AND NOTES SEE PAGE 2F

CABLING:
HEAVY LINES REDUCE
USE OF MULTIPLE LINES

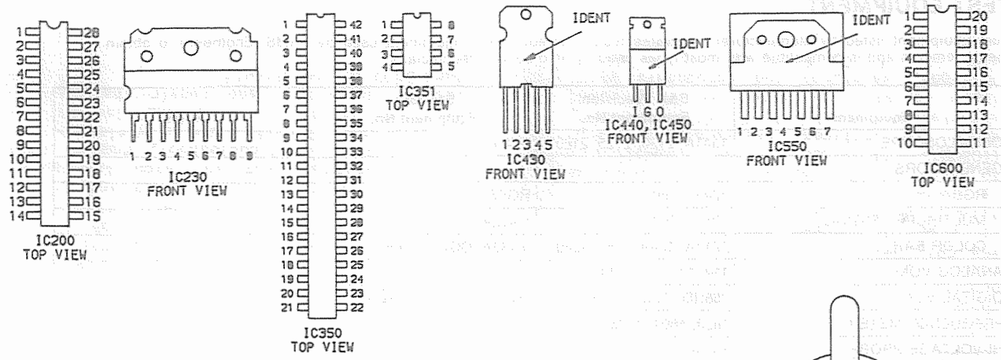


A PHOTOFACT STANDARD NOTATION SCHEMATIC WITH CIRCUITRACE

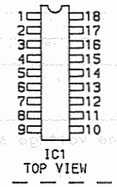
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POWER SUPPLY

E



REMOTE TRANSMITTER
T089AG-MA01 & PH01 VERSIONS



REMOTE TRANSMITTER
T238AG-MA01



For SAFETY use only equivalent replacement part, see parts list.

- Circuitry not used in some versions
- Circuitry used in some versions
- See Parts List
- Nominal value
- Ground
- Chassis
- Common tie point

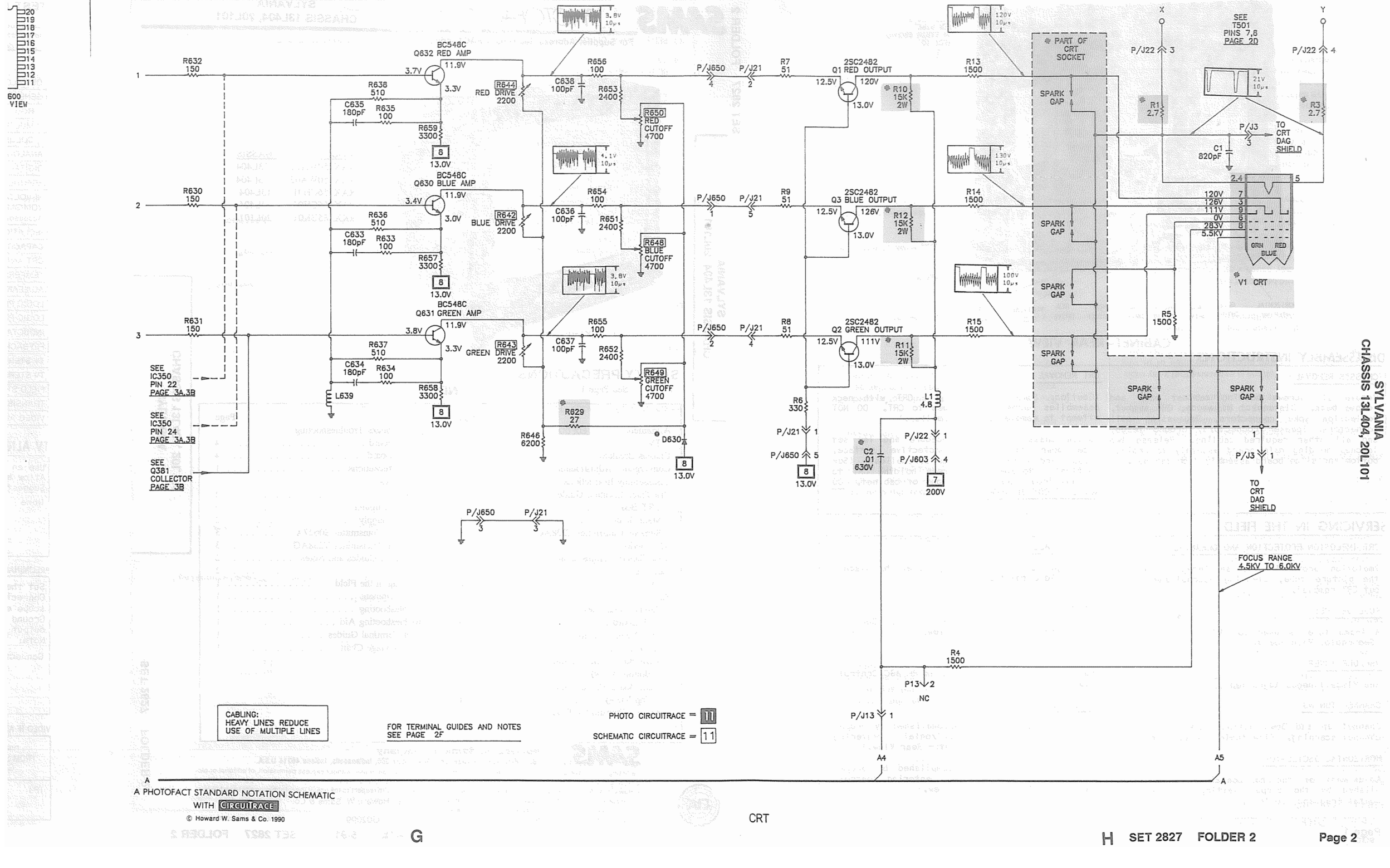
Waveforms and voltages are taken from ground, unless noted otherwise.
Waveforms: triggered scope, keyed rainbow generator.
Item numbers in rectangles appear in the alignment/adjustment instructions.
Supply voltage maintained as shown at input.
Voltages measured with digital meter, no signal.
Controls adjusted for normal operation.
Terminal identification may not be found on unit.
Capacitors are 50 volts or less.
5% or greater unless noted.
Electrolytic capacitors are 50 volts or less,
20% or greater unless noted.
Resistors are 1/2W or less,
5% or greater unless noted.
Value in ( ) used in some versions.
Measurements with switching as shown, unless noted.

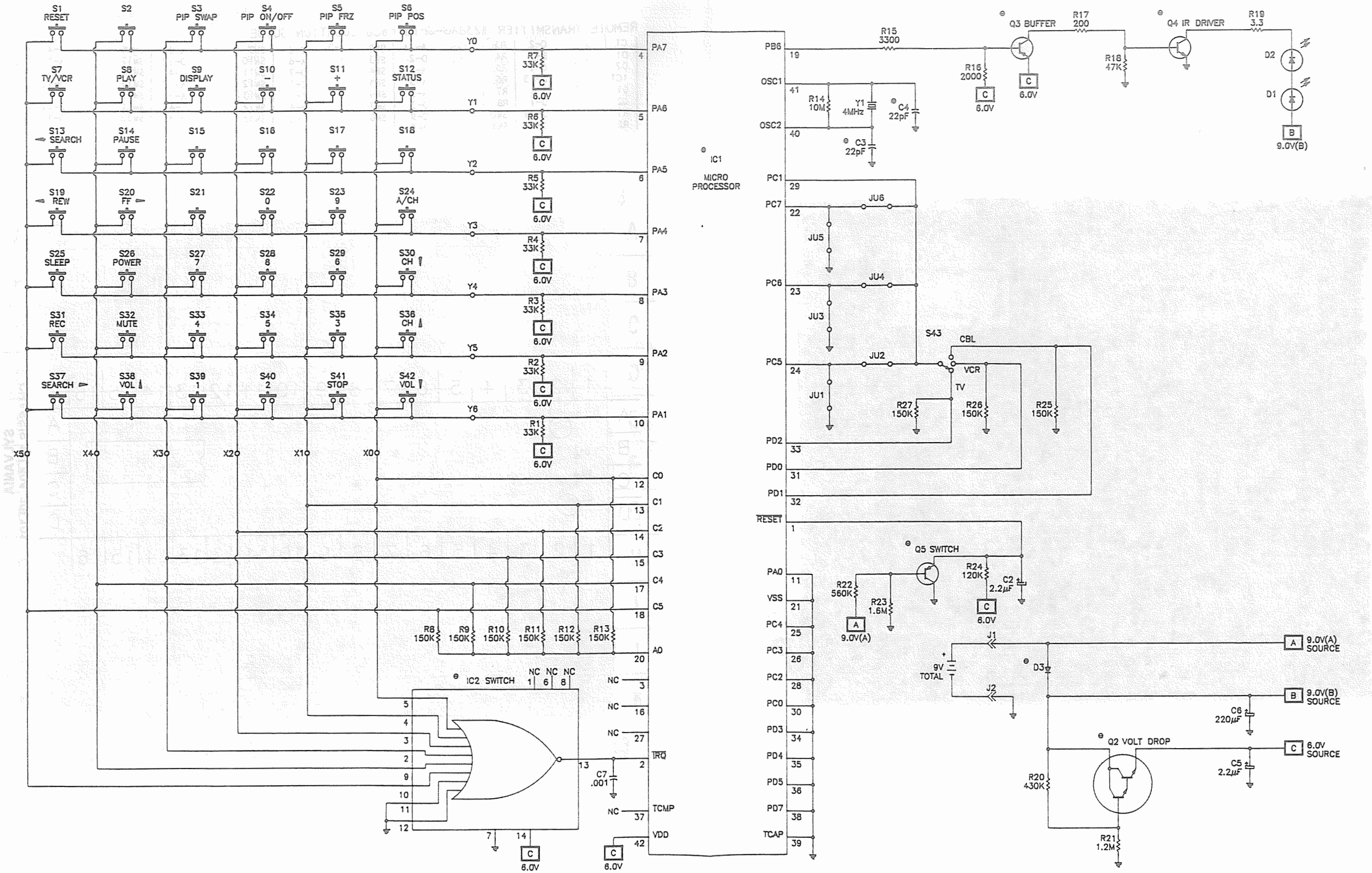
TERMINAL GUIDES AND NOTES

F

SEE IC350 PIN 22 PAGE 3A,3B
SEE IC350 PIN 24 PAGE 3A,3B
SEE Q381 COLLECTOR PAGE 3B

©


600 VIEW
BC548C Q632 RED AMP
BC548C Q630 BLUE AMP
BC548C Q631 GREEN AMP
2SC2482 Q1 RED OUTPUT
2SC2482 Q3 BLUE OUTPUT
2SC2482 Q2 GREEN OUTPUT
R632 150
R630 150
R631 150
R638 510
R636 510
R637 510
C635 180pF
R635 100
C633 180pF
R633 100
C634 180pF
R634 100
L639
R659 3300
R657 3300
R658 3300
R644 DRIVE 2200
R642 DRIVE 2200
R643 DRIVE 2200
RED
BLUE
GREEN
R656 100
R654 100
R655 100
C638 100pF
C636 100pF
C637 100pF
R653 2400
R651 2400
R652 2400
R650 RED CUTOFF 4700
R648 BLUE CUTOFF 4700
R649 GREEN CUTOFF 4700
R629 27
R646 6200
D630
P/J650
P/J21
R7 51
R9 51
R8 51
R13 1500
R14 1500
R15 1500
R10 15K 2W
R12 15K 2W
R11 15K 2W
R6 330
L1 4.8
C2 .01 630V
R4 1500
R5 1500
R1 2.7
R3 2.7
C1 820pF
P/J3
P/J22
P/J603
P13
P/J13
PART OF CRT SOCKET
SPARK GAP
V1 CRT
GRN
RED
BLUE
120V
126V
111V
0V
283V
5.5KV
NC
SEE T501 PINS 7,8 PAGE 2D
TO CRT DAG SHIELD
SEE IC350 PIN 22 PAGE 3A,3B
SEE IC350 PIN 24 PAGE 3A,3B
SEE Q381 COLLECTOR PAGE 3B
FOCUS RANGE 4.5KV TO 6.0KV
CABLING: HEAVY LINES REDUCE USE OF MULTIPLE LINES
FOR TERMINAL GUIDES AND NOTES SEE PAGE 2F
PHOTO CIRCUITRACE = 11
SCHEMATIC CIRCUITRACE = 11
A PHOTOFACT STANDARD NOTATION SCHEMATIC WITH CIRCUITRACE
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SYLVANIA CHASSIS 13L404, 20L101
CRT
SET 2827 FOLDER 2
Page 2



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A

B

PHOTO CIRCUITRACE = 11

SCHEMATIC CIRCUITRACE = 11

FOR TERMINAL GUIDES AND NOTES SEE PAGE 2F

TUNERS NOT INCLUDED IN THIS COVERAGE.

RC RECEIVER
R348 1000
C324 .001
C323 .001
L301
USED IN REMOTE MODELS ONLY
3 5.0V

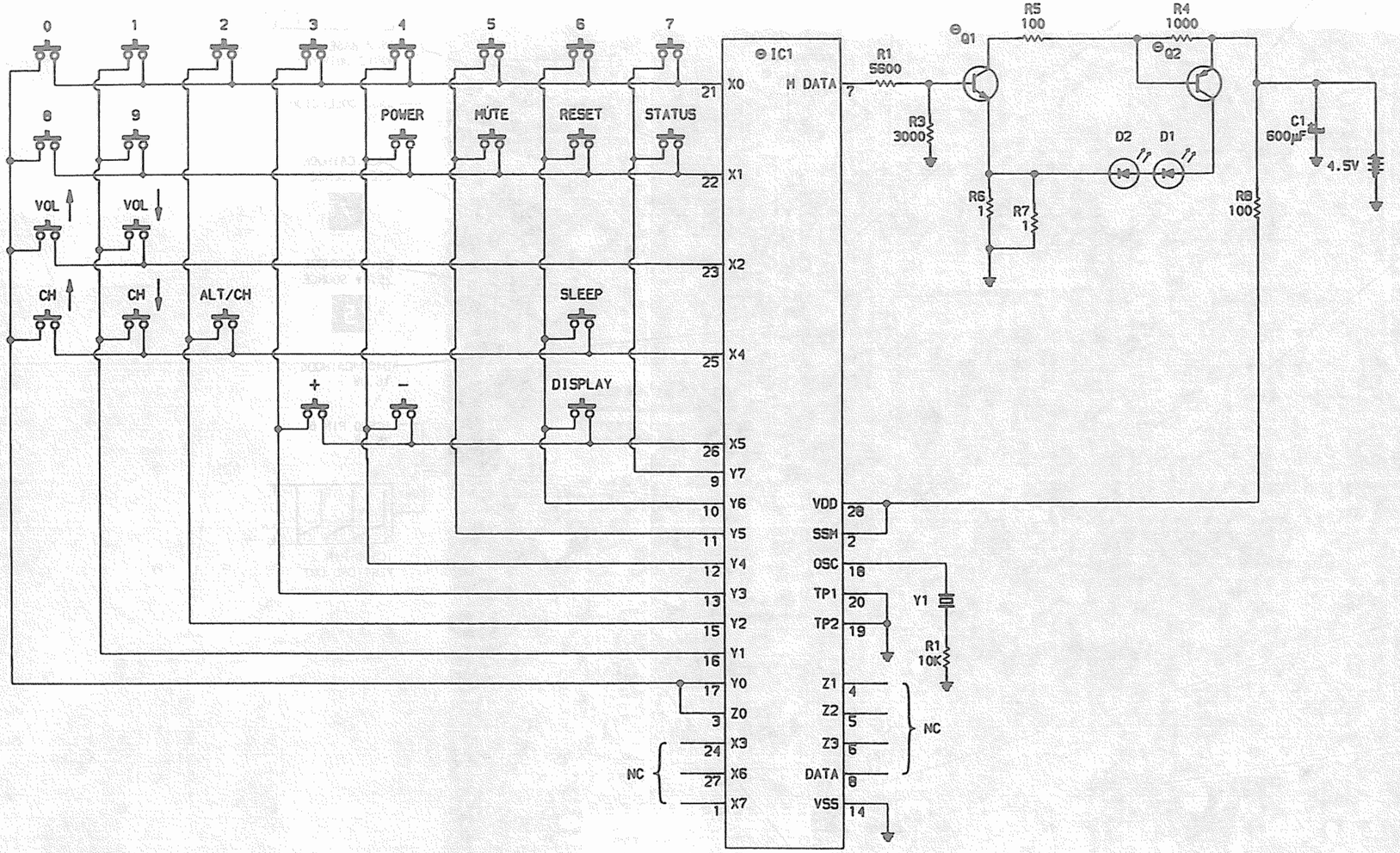
IR INPUT 5.2V 35
SEE IC200 PIN 18 PAGE 2A
AFT 2.6V 9
SEE Q210 EMITTER PAGE 2A
COIN 1.0V 16
Y320 4MHz
R321 820K
C320 27pF NPO
C321 27pF NPO
OSC 2.4V 31
OSC 2.5V 32
USED IN MOTEL MODELS ONLY
NC .2V 1
NC 0V 14
NC .1V 18
NC 0V 8
R314 10
NC .2V 41
TO 14 UHF/VHF TUNER
R311 10
C311 100pF NPO
R310 10
SC .01V 2
C310 100pF NPO
R320 47K
SDC .03V 37
SDA 4.9V 40
IC351 MEMORY
D1 3
D0 4
CS 1
R319 3300
NC 0V 6
NC 7
5 8
4 4.9V
R318 47K
4 4.9V
CHIP SEL .03V 36
NC 0V 20
NC 34
NC 38
.2V
TMP47C634N-2689
IC350
MICROCOMPUTER
R302 1000
I/O 4.9V 10
R305 1000
I/O .1V 11
R304 1000
I/O 4.9V 12
R303 1000
IN 4.5V 13
C302 100pF
C305 100pF
C304 100pF
C303 100pF
R307 10K
3 5.0V
20" MODEL
S70A PWR
S70 PWR
S71 STATUS
S72 +
S73 CHAN UP
S74 -
S75 DISPLAY
S76 VOL UP
S77 VOL DN
S78 CHAN DN
S79
FACTORY USE ONLY
BC548C Q338 VERT PULSE BUFFER
SEE IC550 PIN 7 PAGE 2A
R337 10K
.06V
R338 330
4.7V
R339 4700
3 5.0V
VERT 27
VERT 15
BC548C Q327 HORIZ DELAY BUFFER
SEE IC200 PIN 27 PAGE 2A
R327 82K
1.1V
R335 4700
4.5V
R326 47K (100K) 2%
3 5.0V
A001 VERSION ONLY
Q325 HORIZ PULSE INVERTER
4.5V
4.4V
C328 .001 (330pF) 100V
LATE PRODUCTION MODELS ONLY
R325 10K
3 5.0V
R325 1000
HORIZ 4.4V 26

SCL .1V 39
R312 10
R336 3300
4 4.9V
C312 100pF NPO
TO 13 UHF/VHF TUNER
PICTURE 1.5V 5
R324 24K
R323 33K
R322 15K
C617 10µF
8 13.0V
SEE IC600 PIN 6 PAGE 2C
COLOR .1V 4
R346 27K
R341 91K
R345 47K
C616 2.2µF
SEE IC600 PIN 5 PAGE 2C
SHARPNESS 2.3V 7
R355 10K
R356 68K
C355 47µF
SEE IC600 PIN 15 PAGE 2B,2C
TINT 2.6V 6
R352 22K
R353 43K
R351 16K
C621 22µF
SEE IC600 PIN 17 PAGE 2C
BRIGHT 2.7V 3
R350 22K
R347 33K
R342 68K
R354 11K
C618 22µF
SEE IC600 PIN 9 PAGE 2C
ON/OFF 4.6V 19
SEE Q240 BASE PAGE 2B
R308 100
R306 2000
C306 .001
3 5.0V
R433 10K
1.3V
D414
D433
2SC2482 Q401 POWER ON/OFF
1.0V
.7V
C430 10µF
R432 2400 2W
R429 1800 2W
D406
4400
K400 POWER
2 85.0V
SEE Q410 COLLECTOR PAGE 2B
VOL .4V 2
R330 7500
C330 4.7µF
R331 2940 1%
R332 9100 2%
8 13.0V
BC548C Q330 VOLUME BUFFER
1.6V
1.1V
4 4.9V
R333 1000
SEE IC200 PIN 11 PAGE 2B
GND 34
USED IN 20" MODEL
Q380 RED BUFFER
R386 150
C380 22pF NPO
R380 10K
R372 10K
3 5.0V
2.6V
2.5V
R381 240
1.9V
R373 1200
R377 3000
8 13.0V
D380
SEE IC600 PIN 12 PAGE 2G
RED .1V 22
NC
FAST BLANKING .05V 25
R340 2700
R334 2700
SEE IC600 PIN 11 PAGE 2D
VCC 42
4 4.9V
3 5.0V
C380 22pF
USED IN 20" MODEL
Q382 BLUE BUFFER
R388 150
C382 22pF NPO
R384 10K
R374 10K
3 5.0V
2.6V
2.5V
R385 240
1.9V
R375 1200
R379 3000
8 13.0V
D382
SEE IC600 PIN 14 PAGE 2G
BLUE .1V 24
NC
ON SCREEN DISPLAY COLOR SELECT 5.1V 17
USED IN 13" MODELS ONLY
USED IN 20" MODEL ONLY
R329 15K
R328 15K
3 5.0V
OS CLK 2.5V 28
L357 3.7
OS CLK 2.5V 29
C358 15pF NPO
C357 15pF NPO
BC548C Q381 GREEN BUFFER
SEE IC600 PIN 13 PAGE 2G
GREEN 2.6V 23
R387 150
C381 22pF NPO
R382 10K
R370 10K
3 5.0V
2.6V
2.5V
R383 240
1.9V
R371 1200
8 13.0V
R378 2000 (3000)
D381
RESET 4.6V 33
R313 4700
SEE Q411 COLLECTOR PAGE 2E
C314 560pF
GND 21
GND 30

A PHOTOFACT STANDARD NOTATION SCHEMATIC WITH CIRCUITRACE

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MICROCOMPUTER



A PHOTOFACT STANDARD NOTATION SCHEMATIC

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REMOTE TRANSMITTER T238AG

MISCELLANEOUS ADJUSTMENTS

PRETUNING

NOTE: All procedures require an antenna connected and power applied to the set. Select TV/CATV. Use buttons on receiver front panel or remote transmitter.

Add Channel

1. Press the Display button until the +ADD -DELETE display is on screen.
2. Select channel.
3. Press the +ADD button.
4. Repeat step two and three to add other channels.
5. Press the Status button to end process.

Delete Channel
1. Press the Display button until the +ADD -DELETE display is on screen.
2. Select channel.
3. Press the -DELETE button.
4. Repeat step two and three to erase other channels.
5. Press the Status button to end process.

Sleep
1. Press the Display button until the Set Sleep Timer display is on screen.
2. Press the + Button. Unit can be set to turn off after 30, 60, 90, or 120 minutes by pressing the + Button.

This set employs Digital Customer Controls, use the following control settings for all adjustments unless otherwise indicated: Picture, Brightness, Color, Tint, and Sharpness to Normal Viewing levels.

B+ CHECK

Tune in a picture. Set Brightness, Color, and Picture Controls to MINIMUM. Connect a Digital DC Voltmeter to TP4 (IC430 Pin 4), and Ground. B+ should read 130VDC ± 1VDC.

HIGH VOLTAGE CHECK

Tune in a picture. Set Brightness, Color, and Picture to MINIMUM. Connect a High Voltage Probe to CRT Anode High Voltage should read 24.0KV to 25.5KV. High Voltage must never exceed 25.5KV.

RF AGC ADJUSTMENT

Tune in a picture. Adjust RF AGC Control (R206) Clockwise until snow appears in picture, then reverse direction until snow disappears.

VERTICAL CENTER/SIZE ADJUSTMENT

Tune in a picture. Adjust Vertical Size Control (R559) to reduce size of raster until top and bottom edge are visible. Adjust Vertical Center Control (R545) to provide best vertical centering. Readjust Vertical size control to provide slight overscan without stretching picture.

HORIZONTAL FREQUENCY ADJUSTMENT

Tune in a picture. Set Horizontal Frequency Control (R522) fully Counterclockwise. Adjust Horizontal Frequency Control Clockwise until picture locks in.

HORIZONTAL CENTERING ADJUSTMENT

Tune in a picture. Adjust Horizontal Centering Control (R524) for best horizontal centering.

COLOR PURITY ADJUSTMENT

Operate the set for 15 minutes. Tune in a purity, or crosshatch pattern. Use a degaussing coil to demagnetize the CRT and Mounting brackets. Set Picture, Color, Blue CUTOFF (R648), and Green CUTOFF (R649) to MINIMUM. Set Brightness for a visible raster, Red CUTOFF Control (R650) for a red raster. Loosen the Deflection Yoke clampscrew and slide the Deflection Yoke (L500) backward. Rotate and spread the tabs of the purity magnets until the red band is centered on the screen. Move the Deflection Yoke forward until a uniform red screen is obtained.

GRAY SCALE ADJUSTMENT

Tune in a Crosshatch pattern. Set Brightness, Picture, Color, and Screen (VR400B) to MINI-MUM. Set Red (R650), Blue (R648), Green (R649) CUTOFF Controls, and Red (R644), Green (R643) and Blue (R642) Drive Controls to Mid-range. Remove power from set, remove Plug J10, restore power. Adjust Screen Control until a faint line of one color just appears. Adjust the two remaining CUTOFF Controls to obtain a dim white line. Remove power from set, replace Plug J10, restore power. Set Brightness and Picture Controls to Maximum. Adjust the Drive Controls for best white in highlight areas. Check tracking at low and high brightness. If necessary, retouch Controls as needed.

CONVERGENCE ADJUSTMENTS

Operate the set for 15 minutes. Connect a color bar generator to antenna terminals and tune in a dot pattern. Loosen locking ring. Adjust 4 pole magnets to converge the red and blue dots at the center of the screen. Adjust the 6 pole magnets to converge the red/blue dots over the green dots at the center of the screen. NOTE: Rotate the two tabs of each set of magnets equally and opposite to converge vertically and rotate both tabs in the same direction to converge horizontally. Four and six pole magnets interact, repeat adjustment until center convergence is correct. Tighten locking ring. Tune in a crosshatch pattern. Remove rubber wedges between the Deflection Yoke and the CRT. Tilt Deflection Yoke (L500) up or down to converge the Vertical lines at the top and bottom of the screen and the Horizontal lines at the right and left sides of the screen. Tilt the Deflection Yoke right and left to converge the Horizontal lines at the top and bottom of the

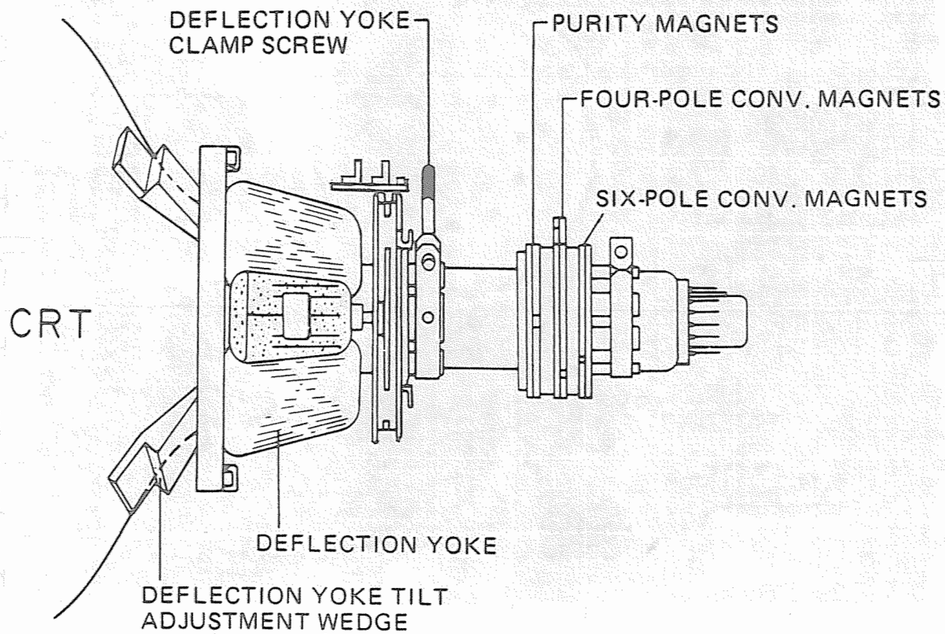
MISCELLANEOUS ADJUSTMENTS (CONTINUED)

screen and the Vertical lines at the right and left sides of the screen. Repeat convergence procedure if necessary to obtain the best overall convergence. Replace the rubber wedges. Tighten Deflection Yoke Clampscrew.

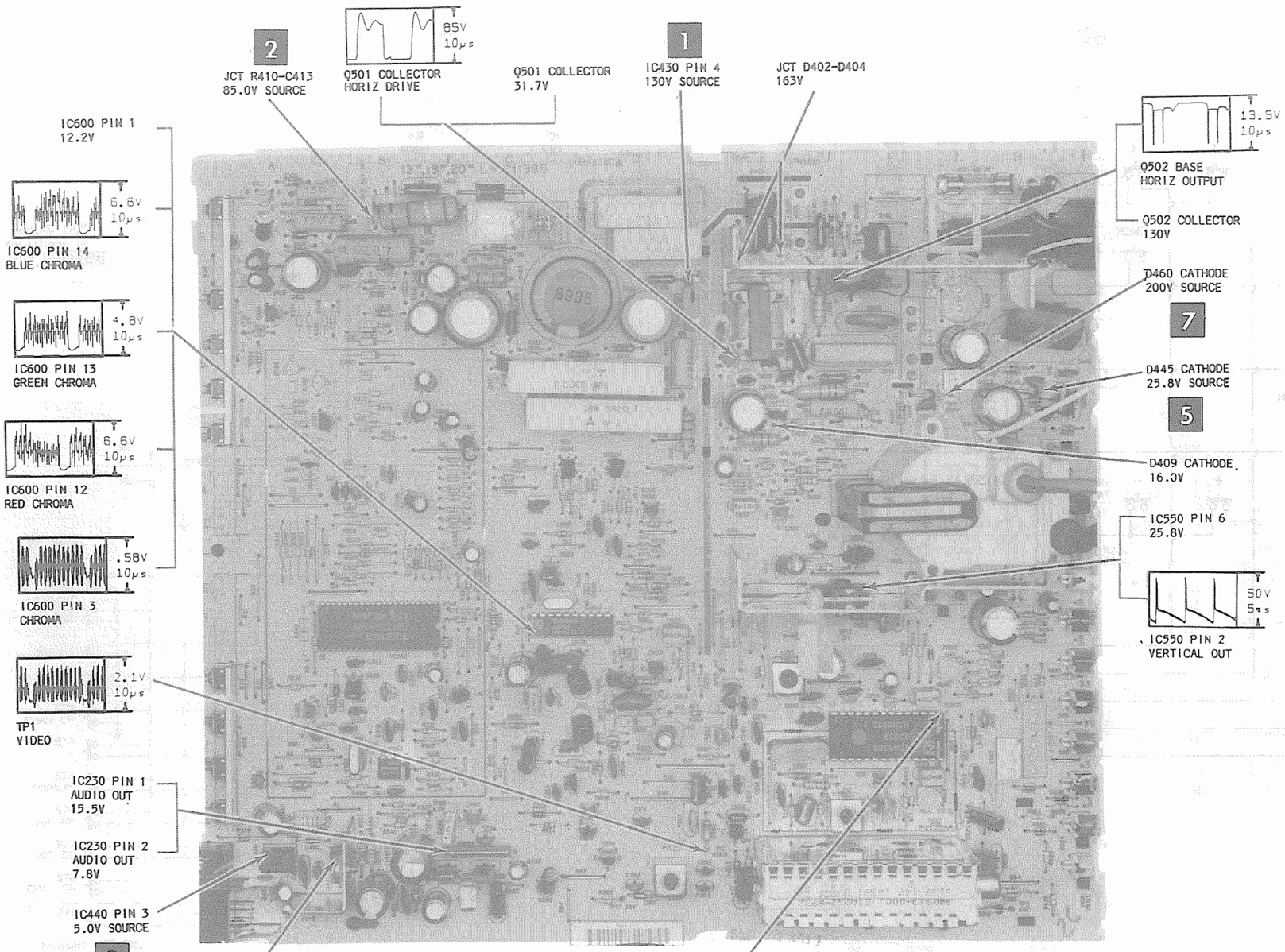
MEMORY RESET VALUES ADJUSTMENT

This adjustment should be performed if the memory IC is replaced. Remote Transmitter must be used for this adjustment. Tune in a picture. Temporarily short IC350 Pin 11 to Pin 12. Factory mode will display on screen. In the Yellow portion of the display the left most number indicates the channel being viewed, the center indicates the test mode "C" and the number (91 to 95) is the DAC to be modified. To select the DAC to modify press 9 and the DAC number (1 to 5). Use the + or - buttons to alter setting. Press the FRONT PANEL POWER BUTTON to store new values.

DAC	FUNCTION	SET TO
91	Brightness	31
92	Picture	47
93	Color	31
94	Tint	31
95	Sharpness	31



CRT NECK ASSEMBLY

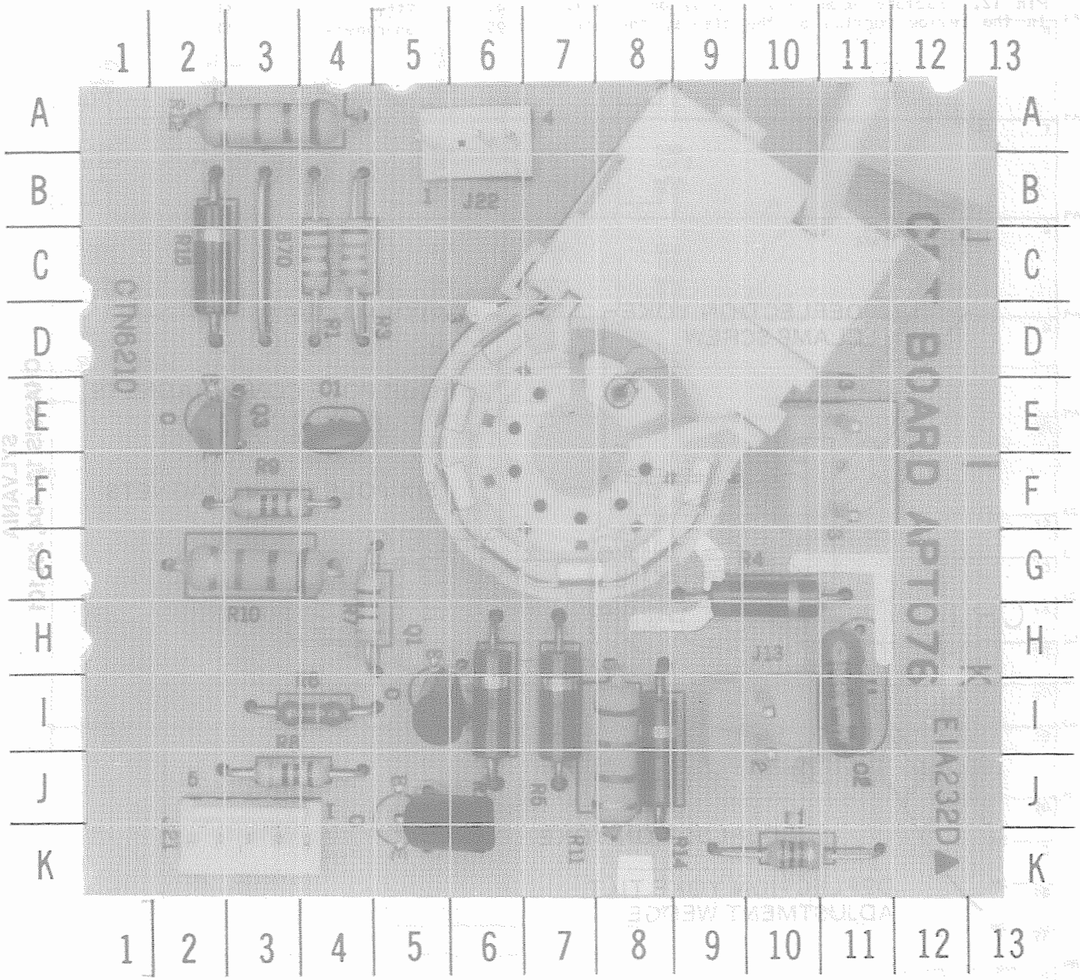


NOTE: ARROWS ON IC'S INDICATE PIN 1 UNLESS NOTED

MAIN BOARD

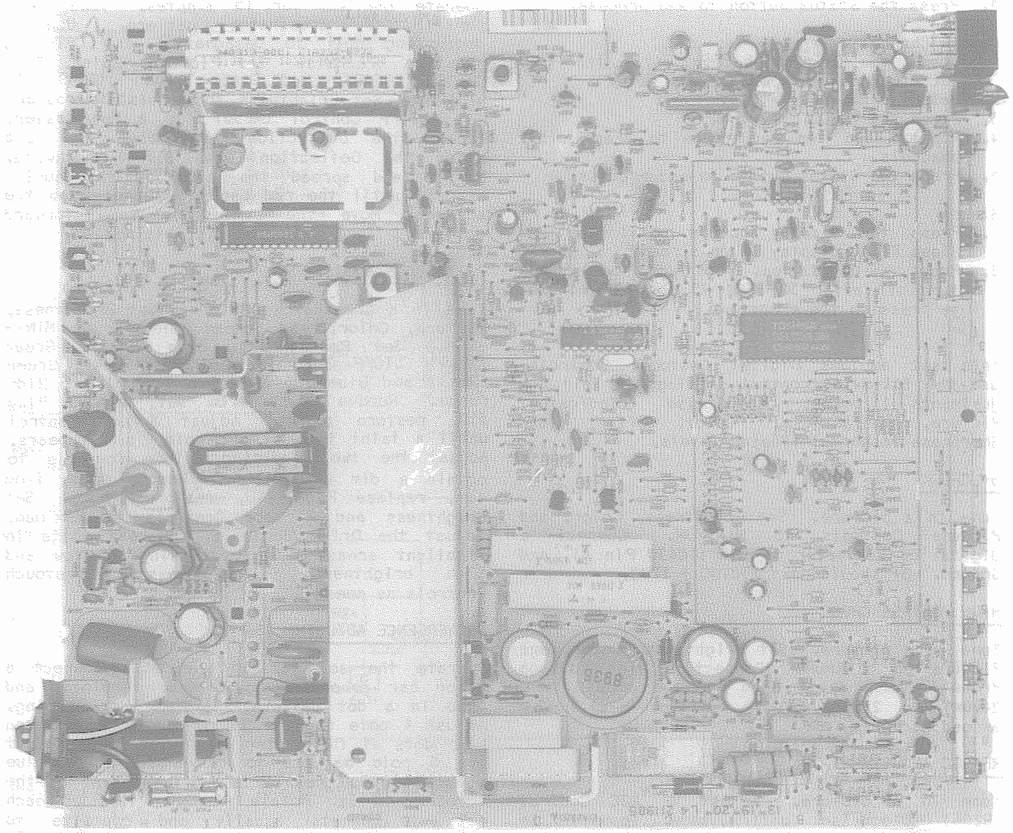
CRT BOARD-GridTrace LOCATION GUIDE

C01	E-4	Q03	E-2	R01	C-4	R07	H-4	R12	A-3
C02	I-11	J3	F-11	R03	C-4	R08	J-3	R13	I-6
L01	K-10	J13	I-10	R04	G-10	R09	F-3	R14	I-8
Q01	I-5	J21	F-3	R05	I-7	R10	I-6	R15	C-2
Q02	J-5	J22	A-6	R06	G-3	R11	J-8		

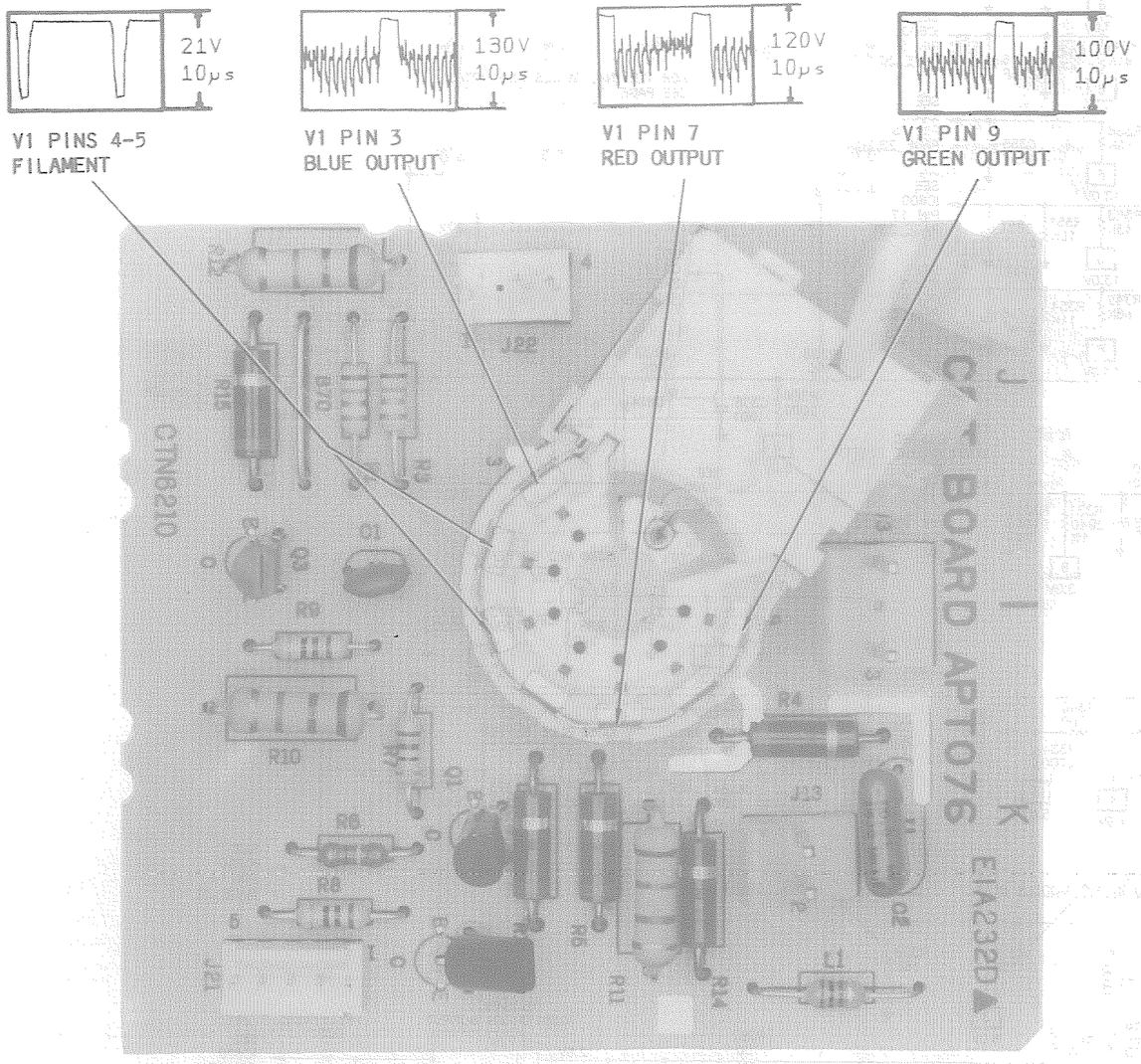


A Howard W. Sams GRIDTRACE™ Photo

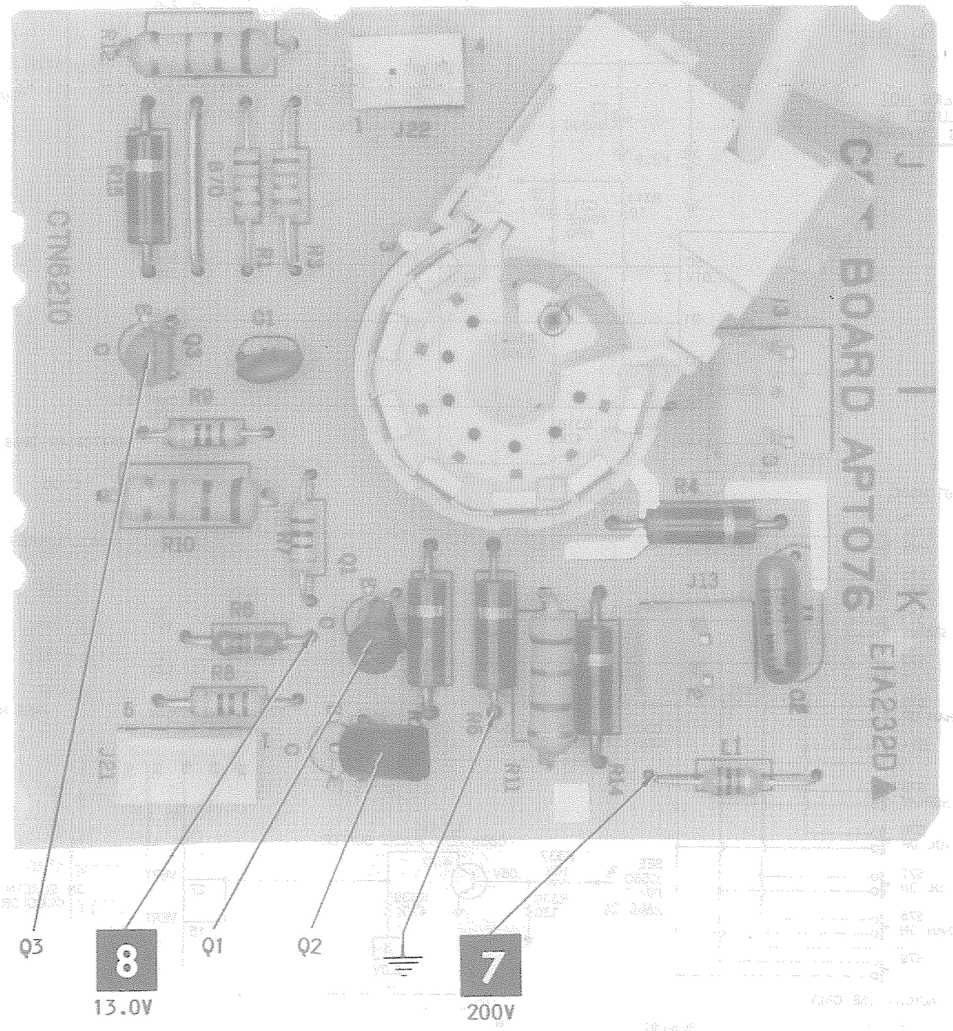
CRT BOARD



MAIN BOARD-SHIELD LOCATION (TOP VIEW)



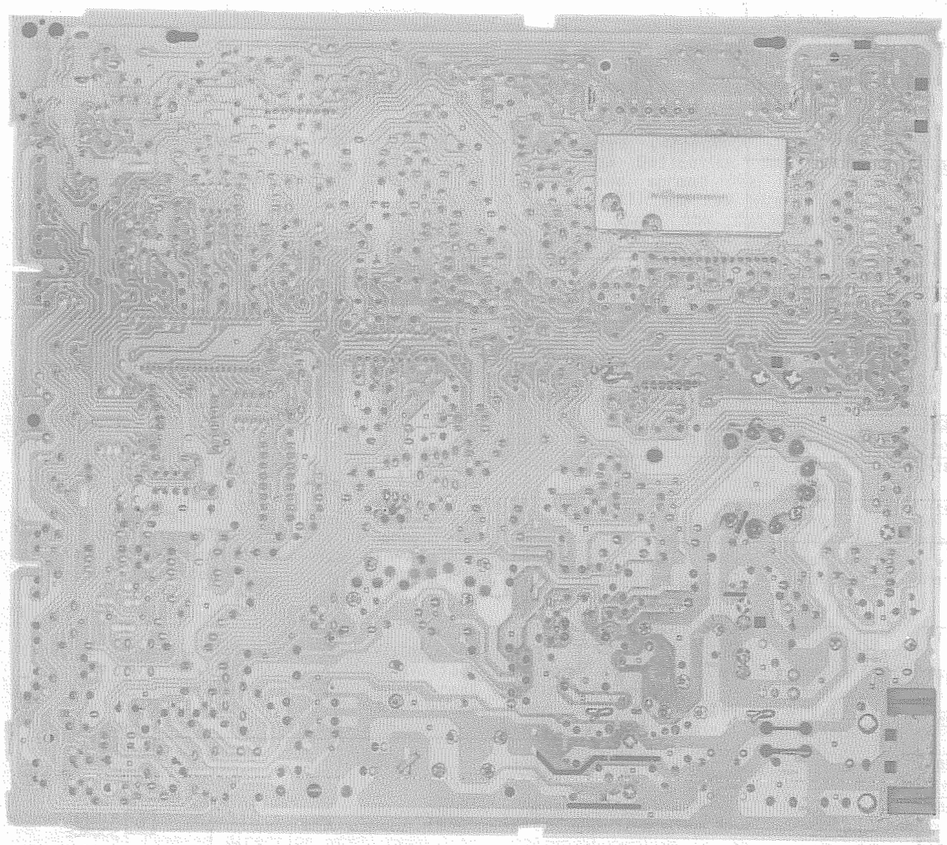
A Howard W. Sams QUICK-CHECKS™ Photo CRT BOARD

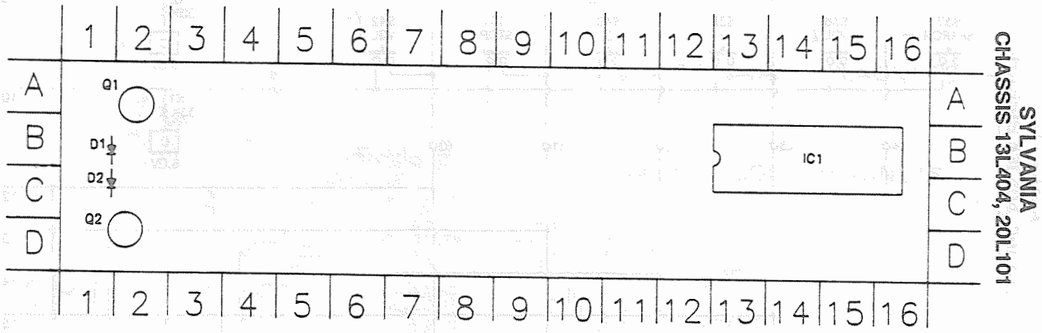


A Howard W. Sams CIRCUITRACE Photo CRT BOARD

REMOTE TRANSMITTER T238AG-GridTrace LOCATION GUIDE

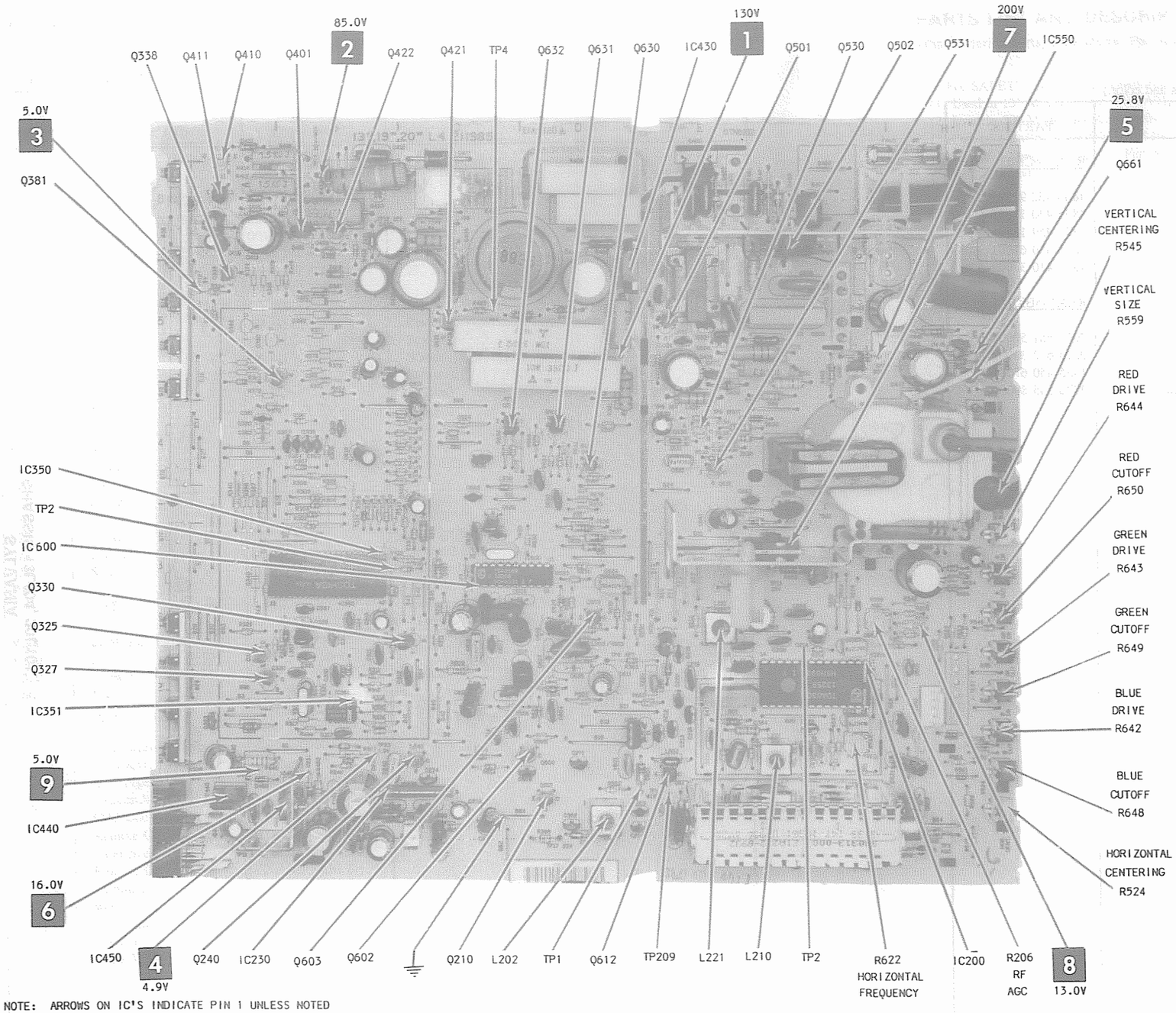
C1	C-2	R3	A-14	SW2	C-6	SW9	A-8	SW16	A-4
D1	B-1	R4	D-2	SW3	A-6	SW10	D-12	SW17	A-5
D2	C-1	R5	A-1	SW4	D-7	SW11	C-4	SW18	C-5
IC1	C-13	R6	A-3	SW5	C-7	SW12	A-12	SW19	D-10
Q1	A-1	R7	A-3	SW6	A-7	SW13	A-10	SW20	A-11
Q2	D-1	R8	D-3	SW7	D-8	SW14	D-4	SW21	C-11
R1	D-14	SW0	C-9	SW8	C-8	SW15	D-5	SW22	D-11
R2	D-13	SW1	D-6						



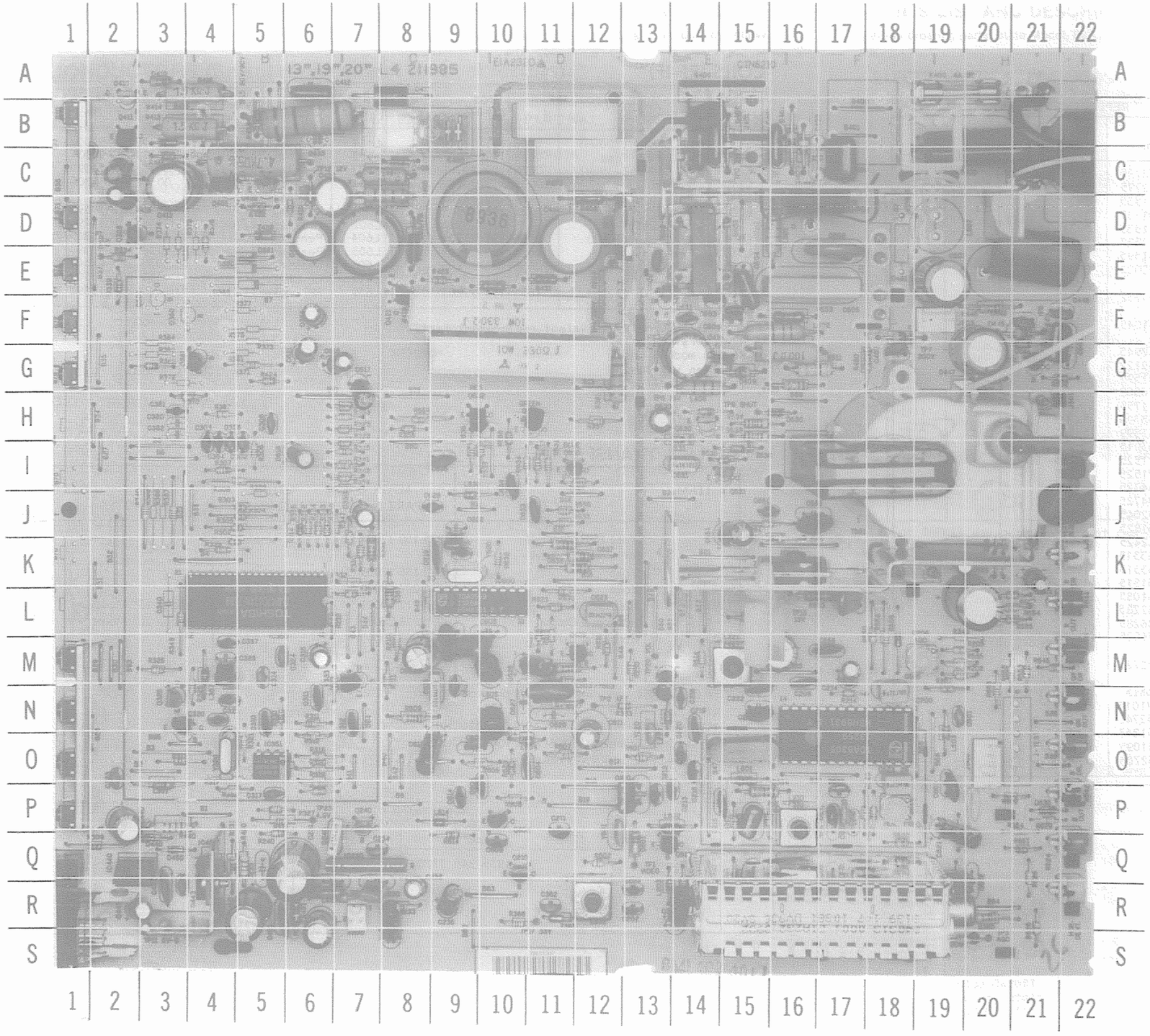


SYLVANIA
CHASSIS 13L404, 20L101

MAIN BOARD-SHIELD LOCATION (BOTTOM VIEW)



NOTE: ARROWS ON IC'S INDICATE PIN 1 UNLESS NOTED



A Howard W. Sams GRIDTRACE™ Photo

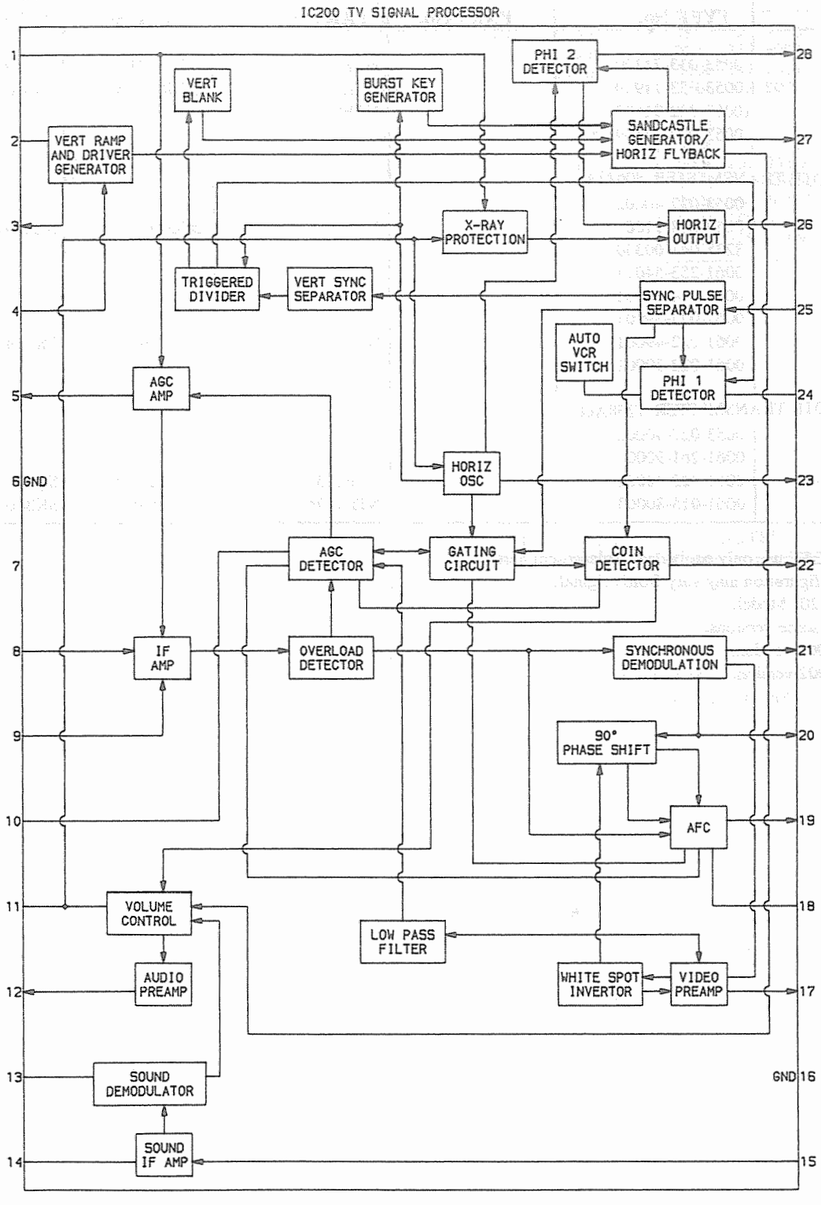
MAIN BOARD

MAIN BO

C201
C202
C203
C206
C208
C210
C211
C212
C213
C214
C215
C216
C218
C219
C220
C221
C222
C230
C231
C232
C234
C235
C236
C237
C240
C241
C302
C303
C304
C305
C306
C310
C311
C312
C314
C315
C317
C320
C310
C311
C312
C314
C315
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C323
C324
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C367
C381
C401
C402
C403
C404
C405
C412
C413
C415
C416
C417
C419
C421

ARD-GridTrace LOCATION GUIDE

R-14	C430	C-6	D661	G-21	R303	J-3	R419	F-8	R628	O-10
R-13	C431	E-11	F400	A-19	R304	J-5	R420	G-16	R629	P-21
R-13	C432	D-7	F401	D-8	R305	J-5	R421	C-5	R630	C-11
M-16	C440	P-2	IC200	O-17	R306	I-5	R422	D-5	R631	I-11
S-17	C441	R-2	IC230	Q-7	R307	I-4	R423	E-9	R632	K-10
P-13	C442	R-3	IC350	L-5	R308	I-5	R424	G-5	R633	J-11
Q-11	C443	R-3	IC351	O-5	R309	Q-2	R429	C-8	R634	I-10
P-13	C445	F-21	IC430	D-13	R310	N-5	R431	E-11	R635	I-9
P-13	C446	F-22	IC440	Q-2	R311	N-6	R432	C-8	R636	J-11
M-17	C447	G-22	IC450	R-4	R312	O-6	R433	B-5	R637	I-11
P-14	C450	R-5	IC550	K-16	R313	O-3	R434	E-10	R638	I-9
N-14	C451	R-5	IC600	L-10	R316	P-5	R436	E-10	R642	C-22
O-14	C460	G-18	IR90	R-1	R318	O-6	R438	F-10	R643	N-22
N-14	C461	E-19	J11	I-18	R319	P-6	R439	G-10	R644	L-22
O-14	C465	M-16	J200	R-7	R320	O-6	R440	P-3	R645	M-21
N-15	C550	J-15	J501	H-21	R321	O-5	R445	F-20	R649	O-22
N-15	C551	J-15	J600	O-20	R322	I-7	R450	Q-5	R650	M-22
R-7	C552	L-17	J603	F-19	R323	H-7	R451	R-6	R651	Q-21
Q-6	C553	L-20	K400	B-8	R324	J-6	R460	G-18	R652	P-20
R-8	C554	K-21	L202	R-12	R325	M-3	R465	L-15	R653	M-21
Q-7	C555	M-17	L203	R-13	R326	N-4	R499	A-3	R654	Q-21
R-9	C556	J-16	L210	P-16	R327	P-5	R501	F-13	R655	N-20
R-6	C558	K-19	L220	N-13	R329	K-4	R502	G-11	R656	M-21
R-6	C565	N-18	L221	M-15	R330	K-7	R503	F-16	R657	I-12
Q-7	C601	Q-13	L301	P-3	R331	K-7	R504	F-12	R658	H-10
M-13	C602	R-19	L357	M-4	R332	K-7	R506	G-16	R659	H-8
I-5	C603	P-19	L365	R-3	R332	M-6	R511	G-16	R662	G-22
I-4	C604	O-10	L406	G-14	R334	E-2	R514	M-14	R663	G-20
I-4	C605	P-10	L501	D-16	R335	P-5	R520	Q-18	S70	B-1
I-4	C606	O-9	L502	F-16	R336	P-6	R521	P-18	S71	D-1
H-5	C607	N-9	L503	N-19	R337	H-12	R522	P-18	S72	F-1
N-5	C608	M-9	L601	O-15	R338	C-2	R523	P-18	S73	B-1
N-6	C609	Q-10	L602	P-13	R339	E-2	R524	R-22	S74	E-1
N-7	C610	M-8	L603	N-11	R340	J-3	R525	Q-21	S75	G-1
M-5	C611	O-16	L604	O-10	R341	I-7	R526	R-21	S76	N-1
M-6	C612	O-9	L605	N-10	R342	I-7	R530	J-22	S77	M-1
P-5	C613	M-9	L606	M-10	R345	I-7	R531	J-22	S78	O-1
N-4	C614	R-9	L614	P-9	R346	J-6	R533	H-15	T501	I-19
N-5	C615	N-9	Q210	Q-10	R347	J-7	R534	H-15	T502	E-14
N-6	C616	G-7	Q240	Q-7	R348	L-3	R535	J-22	TP1	Q-13
N-7	C617	G-7	Q325	N-3	R350	J-7	R536	I-15	TP2	L-7
M-5	C618	I-6	Q327	N-3	R351	H-7	R545	I-22	TP4	E-9
M-6	C619	K-9	Q330	M-7	R352	J-6	R550	M-18	TP209	Q-13
P-5	C620	K-9	Q338	D-2	R353	H-7	R551	K-15	TP210	N-17
N-4	C621	G-6	Q381	G-4	R354	I-7	R553	L-16	TUNER	K-17
N-4	C622	K-9	Q401	C-4	R355	J-6	R554	L-21	Y200	O-15
O-2	C623	L-12	Q410	B-2	R356	H-8	R555	L-21	Y220	O-13
M-5	C625	M-11	Q411	B-2	R360	R-20	R556	M-18	Y320	O-4
M-4	C628	J-9	Q421	F-8	R361	G-12	R557	L-20	Y601	Q-12
J-7	C633	J-11	Q501	E-13	R362	G-12	R558	M-18	Y603	N-11
F-6	C634	I-10	Q502	D-16	R363	P-15	R559	K-22	Y620	K-9
M-4	C635	I-9	Q530	H-14	R364	L-7	R560	K-20	Z361	R-11
N-4	C636	K-11	Q531	I-15	R366	R-10	R561	K-20	Z401	B-4
S-16	C637	H-12	Q602	P-10	R369	S-16	R562	J-21	Z410	B-4
R-4	C638	K-12	Q603	M-12	R370	H-4	R565	M-19	Z411	B-3
R-11	D91	Q-1	Q612	Q-13	R371	G-5	R601	O-13	Z421	C-6
P-15	D381	E-5	Q630	I-11	R378	D-4	R602	Q-12	Z422	D-5
P-15	D401	B-16	Q631	H-11	R381	H-3	R603	Q-19	Z530	I-15
R-8	D402	B-15	Q632	H-9	R382	G-3	R604	N-10	Z531	H-14
L-8	D403	B-16	Q661	G-21	R383	H-4	R605	O-10		
H-3	D404	C-14	R201	R-13	R387	J-3	R606	M-11		
B-20	D406	C-8	R205	M-19	R400	B-14	R607	O-11		
B-16	D409	G-15	R208	M-19	R401	D-21	R608	P-10		
B-14	D413	A-8	R209	S-16	R402	B-11	R609	N-8		
B-14	D414	C-4	R210	M-8	R403	C-12	R610	M-12		
D-10	D433	C-5	R212	L-6	R405	D-12	R611	M-11		
A-6	C440	Q-3	R214	N-17	R410	B-6	R612	Q-12		
C-3	D445	F-21	R230	M-13	R413	B-4	R614	Q-9		
D-2	D450	Q-3	R231	Q-13	R414	A-2	R623	L-11		
C-2	D460	F-18	R232	R-7	R415	A-2	R625	M-11		
G-15	D530	H-16	R233	R-8	R416	A-2	R626	L-11		
G-14	D550	K-16	R240	Q-5	R417	C-3	R627	N-11		
D-6	D630	P-21	R302	K-4	R418	D-6				



IC FUNCTIONS

CHASSIS BREAKDOWN

13L404

00EML404	Main Chassis Board Assembly
00EML405	Main Chassis Board Assembly
UV936	Varactor Tuner Module
00APT076	CRT Socket Board Module

20L101

OOEML122	Main Chassis Board Assembly
UV936	Varactor Tuner Module
00APT076	CRT Socket Board Module

PARTS LIST AND DESCRIPTION

When ordering parts, state Model, Part Number, and Description

SEMICONDUCTORS (Select replacement for best results)

	ITEM No.	MFR TYPE No.	MFR PART No.	NTE PART No.	ECG PART No.	TCE PART No.
	D380 (1)	0053-018-11001		NTE177	ECG177	SK9091
	D381	0053-018-11001		NTE177	ECG177	SK9091
	D382 (1)	0053-018-11001		NTE177	ECG177	SK9091
	D401 - D404	BYD14K		NTE125	ECG125	SK3081
	D406	0053-018-11001		NTE177	ECG177	SK9091
	D409	BYV95C		NTE580	ECG580	SK5036
	D410 (2)	0053-024-71001		NTE112	ECG112	SK3089
	D413	BYW95C		NTE580	ECG580	SK5036
	D414	0053-024-71001		NTE112	ECG112	SK3089
	D433	0053-018-11001		NTE177	ECG177	SK9091
	D440	0053-018-11001		NTE177	ECG177	SK9091
	D445	BYV95C		NTE580	ECG580	SK5036
	D450	0053-018-11001		NTE177	ECG177	SK9091
	D460	BYV95C		NTE580	ECG580	SK5036
#	D530	0053-018-11001		NTE177	ECG177	SK9091
	D550	0053-031-01001		NTE580	ECG580	SK5036
	D630	0053-018-11001		NTE177	ECG177	SK9091
	D661	0053-026-11001		NTE552	ECG552	SK9000
	IC200	TDA8305				
	IC230	TDA2611A		NTE1566	ECG1566	SK7726
	IC350	TMP47C634N-2689				
	IC350 (3)	8235-041-04130				
	IC350 (4)	8235-041-04180				
	IC351	NMC9306N				
	IC430	STR30130		NTE1777	ECG1777	
	IC440	0061-247-90001		NTE960	ECG960	SK3591
	IC450	0061-247-90004				
	IC550	LA7830		NTE1773	ECG1773	SK9752
	IC600	TDA3569				
	Q1,2,3	C2482		NTE399	ECG399	SK9352
	Q210	0061-043-49003				
	Q240	0061-043-49003				
	Q325 (4)	C558		NTE159*	ECG159*	SK3466*
	Q327	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q330,38	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q380 (1)	0061-043-59002		NTE123AP*	ECG123AP*	SK3854*
	Q381	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q382 (1)	0061-043-59002		NTE123AP*	ECG123AP*	SK3854*
	Q401	C2482		NTE399	ECG399	SK9352
	Q410	0061-022-49001		NTE199*	ECG199*	SK3245*
	Q411	0061-043-49003				
	Q421	0061-044-30001		NTE48	ECG48	SK4906
	Q422	0061-043-59002		NTE123AP*	ECG123AP*	SK3854*
	Q501	0061-055-49001		NTE399	ECG399	SK9352
	Q502	D1554		NTE2302	ECG2302	SK9422
#	Q530	0061-043-49003				
	Q531	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q602,3	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q612	0061-043-49003				
	Q630,1,2	C548C		NTE123AP*	ECG123AP*	SK3854*
	Q661	0061-043-49003				
	Z361	0053-033-71330		NTE5036A	ECG5036A	SK33A
	Z410	0053-033-71629				
	Z411	0053-033-71479		NTE5009A	ECG5009A	SK4A7

PARTS LIST AND DESCRIPTION (Continued)

When ordering parts, state Model, Part Number, and Description

RESISTORS (Power and Special)

	ITEM No.	RATING	REPLACEMENT DATA MFGR. PART No.	NTE PART No.	
#	R1	2.7 5% 1/2W Metal Film	0022 032 72795	HW2D7	
#	R3	2.7 5% 1/2W Metal Film	0022 032 72795	HW2D7	
#	R10	15K 5% 2W Metal Film	0023 033 61535	2W315	
#	R11	15K 5% 2W Metal Film	0023 033 61535	2W315	
#	R12	15K 5% 2W Metal Film	0023 033 61535	2W315	
	R230	7500 2% 1/8W Carbon Film	0023 031 57522	EW275	
#	R231	15 5% 1/4W Metal Film	0023 026 81595	QW015	
	R233	2000 2% 1/8W Carbon Film	0023 031 52022	EW220	
	R326	47K 5% 1/8W Carbon Film		EW347	
		47K 2% 1/8W Carbon Film	0023 031 54732 (1)	EW247	
		100K 5% 1/8W Carbon Film	0023 031 51045 (2)	EW410	
	R331	29.4K 1% 1/4W Carbon Film	0023 027 52943		
	R332	9100 2% 1/8W Carbon Film	0023 031 59122	EW291	
#	R360	15K 5% 1/4W Metal Film	0023 026 81595	QW315	
#	R400	11.1 PTC Cold	0023 020 70008		
#	R401	4.7M 5% 1/2W Carbon Comp	0023 023 44755	HW547	
#	R402	3.3 10% 5W Wirewound	0024 014 53399	5W3D3	
#	R403	3.3 10% 5W Wirewound	0024 014 53399	5W3D3	
#	R410	390 5% 5W Metal Film	0023 033 83915		
#	R413	1500 10% 3W Metal Film	0023 033 71525	3W215	
#	R414	1500 10% 3W Metal Film	0023 033 71525	3W215	
#	R420	.51 5% 1/2W Metal Film	0023 033 84725	HWD51	
#	R421	4700 5% 5W Metal Film	0023 033 84725		
#	R423	200K 5% 1/8W Carbon Film	0023 031 52045	EW420	
#	R429	1800 5% 2W Metal Film	0023 033 61825	2W218	
#	R432	2400 5% 2W Metal Film	0023 033 62425	2W224	
#	R438	330 5% 10W Wirewound	0024 012 63315	10W133	
#	R439	330 5% 10W Wirewound	0024 012 63315	10W133	
#	R440	150 5% 1W Metal Film	0023 033 51515	1W115	
#	R445	1 5% 1/4W Metal Film	0023 026 81085	QW1D0	
#	R460	2.2 5% 1/4W Metal Film	0023 026 82285	QW2D2	
#	R465	6.2 5% 1/4W Metal Film	0023 026 86285	QW6D2	
#	R504	6800 5% 3W Metal Film	0023 033 76825 (1)	3W268	
		2700 5% 3W Metal Film	0023 033 72725 (2)	3W227	
#	R507	680 5% 1/2W Carbon Film	0023 028 26815	HW168	
#	R506	100 5% 3W Metal Film	0023 033 71015	3W110	
#	R533	2.74K 1% 1/4W Metal Film	0023 027 52742		
#	R534	12.1K 1% 1/4W Metal Film	0023 027 51213		
#	R609	10 5% 1/4W Metal Film	0023 026 81095	QW010	
#	R629	27 5% 1/8W Carbon Film	0023 031 52705	EW027	

For SAFETY use only equivalent replacement part.
(1) B002 versions only.
(2) A001 versions only.

PARTS LIST AND DESCRIPTION (Continued)

When ordering parts, state Model, Part Number, and Description

COILS (RF-IF)

ITEM No.	FUNCTION	MFGR. PART No.
L1	RF Choke (100uH)	0036 181 31019
L202	47.25MHz Trap	0036 179 90007
L203	RF Choke (.39uH)	0036 209 70390 (1)
L210	4.5MHz	0036 179 90009
L220	Peaking (27uH)	0036 209 72705
L221	4.5MHz Discriminator	0036 196 80005
L301	RF Choke (12uH)	0036 181 31209 (2)
L357	Peaking (39uH)	0036 209 73905
L365	RF Choke (12uH)	0036 181 31209
# L400	Line Filter	0036 191 50004 (3)
L406	RF Choke (12uH)	4835 154 97001

ITEM No.	FUNCTION	MFGR. PART No.
L501	Peaking (5.3uH)	4835 154 97002
L502	RF Choke (25uH)	0036 212 90004
L503	Peaking (3.3uH)	0036 209 73399
# L507	Horizontal Linearity	0036 202 80009 (4)
L601	Peaking (1.5uH)	0036 181 31590
L602	Peaking (4.7uH)	0036 181 34795
L604	Peaking (18uH)	0036 209 71809
L605	Peaking (6.8uH)	0036 206 66890
L606	Peaking (15uH)	0036 206 61509
L614	RF Choke (15uH)	0036 209 71509
L639	RF Choke (1uH)	0036 209 71090

For SAFETY use only equivalent replacement part.
(1) A001 versions only.
(2) Remote models only.
(3) Canadian version.
(4) Used in model with 20" CRT.

COILS & TRANSFORMERS

	ITEM No.	FUNCTION	MFGR PART No.	OTHER IDENTIFICATION	NOTES
#	L500	Yoke Horiz 3.78mH 90° Vert 28.2mH	4835 150 17004 (2)		
		Yoke	4835 150 17005 (3)		
		Yoke	0036 207 30003 (4)		
#	T501	Horizontal Output	0036 212 10001	0036 212-1 (1)	Used in 13" Models.
		Horizontal Output	0036 212 20001		Used in 20" Model.
	T502	Horizontal Driver	0032 040 30003		

For SAFETY use only equivalent replacement part.
(1) Number on unit.
(2) Used in Models RAX150WA01, RAX156CH01.
(3) Used in Models RAX150WA02, RAX156CH02.
(4) Used in Model RKX185BK01.

SPEAKER

ITEM No.	TYPE	REPLACEMENT DATA MFGR. PART No.	QUAM PART No.	NOTES
	3" 16 Ohms	4835 240 37002		

PARTS LIST AND DESCRIPTION (Continued)

When ordering parts, state Model, Part Number, and Description

CABINETS & CABINET PARTS (When ordering specify model, chassis and color.)

ITEM	PART No.
Model	RAX150WA01
Cabinet Back	4835 432 97004
Cabinet Front	4835 430 17005
Inlay-Control	4835 459 17003
Lens IR	4835 450 67001
Pushbutton Cluster	4835 410 37001
Model	RAX150WA02
Cabinet Back	4835 432 97004
Cabinet Front	4835 430 17005
Inlay-Control	4835 459 17003
Lens IR	4835 450 67001
Pushbutton Cluster	4835 410 37001
Model	RAX156CH01
Cabinet Back	4835 432 97003
Cabinet Front	4835 430 17003
Inlay-Control	4835 459 17005
Lens IR	4835 450 67001
Pushbutton Cluster	4835 410 37002

ITEM	PART No.
Model	RAX156CH02
Cabinet Back	4835 432 97003
Cabinet Front	4835 430 17003
Inlay-Control	4835 459 17005
Lens IR	4835 450 67001
Pushbutton Cluster	4835 410 37002
Model	RKX185BK01
Cabinet Back	0014 651 50001
Cabinet Front	0014 652 30002
Lens IR	4835 450 67001
Control Button Assembly	0070 545 80006

PARTS LIST AND DESCRIPTION (Continued)

When ordering parts, state Model, Part Number, and Description

SEMICONDUCTORS (Select replacement for best results)

	ITEM No.	MFR TYPE No.	MFR PART No.	NTE PART No.	ECG PART No.	TCE PART No.
	Z421	0053-033-71120		NTE5021A	ECG5021A	SK12A
	Z422	0053-033-71919		NTE5018A	ECG5018A	SK9A1
#	Z530	0053-032-91629		NTE5013T1	ECG5013T1	
	Z531	0053-032-91439				
	REMOTE TRANSMITTER S0627A					
	D1,2	0053-027-40002				
	D3	0053-018-11001		NTE177	ECG177	SK9091
	IC1	8235-041-00330				
	IC2	0061-253-54078				
	Q2	0061-044-39001				
	Q3	0061-022-39001				
	Q4	0061-022-49001		NTE199*	ECG199*	SK3245*
	Q5	0061-022-39001				
	REMOTE TRANSMITTER T238AG					
	D1,2	0053-027-40002				
	IC1	0061-261-90001				
	Q1	0061-022-40001		NTE123AP	ECG123AP	SK3854
	Q2	0061-015-80003		NTE159*	ECG159*	SK3466*

For SAFETY use only equivalent replacement part.
* Lead configuration may vary from original.
(1) Used in 20" Model.
(2) Used in some versions.
(3) Used A001 version.
(4) Used B002 version.

PARTS LIST AND DESCRIPTION (Continued)

When ordering parts, state Model, Part Number, and Description

ELECTROLYTIC CAPACITORS Items Not Listed Are Normally Available At Local Distributors.

CAPACITORS Items Not Listed Are Normally Available At Local Distributors.

	ITEM No.	RATING	MFGR. PART No.
#	C2	.01 630V 20%	0025 081 61030
	C202	4.7pF NPO 50V ±.25pF	0025 084 14797
	C203	10pF NPO 50V	0025 084 11008
	C211	56pF NPO 50V 5%	0025 084 15605
	C212	56pF NPO 50V 5%	0025 084 15605
	C213	56pF NPO 50V 5%	0025 084 15605
	C216	22pF NPO 50V 5%	0025 084 12205
	C218	47pF NPO 50V 5%	0025 084 14705
	C220	27pF NPO 50V 5%	0025 084 12705
	C310	100pF NPO 50V 5%	0025 083 11015
	C311	100pF NPO 50V 5%	0025 084 11015
	C312	100pF NPO 50V 5%	0025 084 11015
	C320	27pF NPO 50V 5%	0025 084 12705
	C321	27pF NPO 50V 5%	0025 084 12705

	ITEM No.	RATING	MFGR. PART No.
	C357	15pF NPO 50V 5%	0025 084 11505
	C358	15pF NPO 50V 5%	0025 084 11505
	C381	22pF NPO 50V 5%	0025 084 12205
	C382	22pF NPO 50V 5%	0025 084 12205
#	C400	.22 125VAC	0025 098 42240(1)
		.1 125VAC	(2)
#	C505	.0039 1.5KV 5%	0025 081 83925
#	C506	.001 2KV 10%	0025 088 50009
#	C507	.27 200V 5%	0025 080 50012(3)
		.36 200V	0025 080 50010
	C603	150pF NPO 50V 5%	0025 084 11515
	C605	47pF NPO 50V 5%	0025 084 14705
	C609	27pF NPO 50V 5%	0025 084 12705
	C614	47pF NPO 50V 5%	0025 084 14705

For SAFETY use only equivalent replacement part.
(1) USA version.
(2) Canadian version.
(3) Model with 20" CRT.

CONTROLS (All wattages 1/2 watt, or less, unless listed)

	ITEM NO.	FUNCTION	RESISTANCE	MFGR. PART NO.	NOTES
	R206	RF AGC	22K	0022 048 02232	# For SAFETY use only equivalent replacement part.
	R522	Horizontal Frequency	10K	0022 048 01032	
	R524	Horizontal	47K	0022 048 34732	(1) Part of Horizontal Output Transformer T501. Part No. 0036 212 10001.
	R545	Vertical Centering	10K	0022 048 31032	
	R559	Vertical Size	110	0022 048 32212	
	R642	Blue Drive	2200	0022 048 32222	
	R643	Green Drive	2200	0022 048 32222	
	R644	Red Drive	2200	0022 048 32222	
	R648	Blue Cutoff	4700	0022 048 34722	
	R649	Green Cutoff	4700	0022 048 34722	
	R650	Red Cutoff	4700	0022 048 34722	
#	VR400A	Focus		(1)	
#	VR40B	Screen		(1)	

MISCELLANEOUS

	ITEM No.	DESCRIPTION	MFGR PART No.	NOTES
	D91	LED	4835 130 8700	Power, Green
#	F400	Fuse 4 Amp @ 125V	0018 152 05400	Slow-Blow
#	F401	Fuse 2 Amp @ 250V		
		Fuse 1.5 Amp @ 125V	0018 169 05150	
#	K400	Power Relay	0016 072 30011	
#	L401	Degaussing Coil	0036 209 40012	13" Models
			0036 209 40007	20" Model
#	P400	AC Cord	0046 140 70009	
	S70	Switch	(1) (4)	Power
		Switch	(2) (5)	Power
	S70A	Switch	(2) (5)	Power
	S71	Switch	(1) (4)	Status
		Switch	(2) (5)	Status
	S72	Switch	(1) (4)	+ (Plus)
		Switch	(2) (5)	+ (Plus)
	S73	Switch	(3)	Channel Up
	S74	Switch	(1) (4)	- (Minus)
		Switch	(2) (5)	- (Minus)
	S75	Switch	(1) (4)	Display
		Switch	(2) (5)	Display
	S76	Switch	(3)	Volume Up
	S77	Switch	(3)	Volume Down
	S78	Switch	(3)	Channel Down
	S79	Switch		Factory Use Only
	V1	CRT	370KSB22	Models: RAX150WA01, RAX156CH01
		CRT	A34JLN60X	Models: RAX150WA02, RAX156CH02
		CRT	A51ADY00	Model: RKX185BK01
	Y200	SAW Filter	0036 300 20002	
	Y220	Filter	0036 179 60006	
	Y320	Crystal	0056 044 40015	4MHz
	Y601	Filter	0036 175 60001	
		Balun	0036 180 50003 (1)	
		Balun	4835 218 27003 (2)	
		Convergence/Purity Assembly	0036 202 60002 (1)	
		Convergence/Purity Assembly	4835 150 27001 (2)	
		CRT Socket	0018 162 70005	
		Remote Control Receiver	0070 542 10001	
		Remote Control Transmitter	T238AG-SA01 (1)	
		Remote Control Transmitter	S0627-AA01 (2)	
		Yoke Wedge	0064 585 20002 (1)	3 used
		Yoke Wedge	4835 535 27001 (2)	3 used

PARTS LIST AND DESCRIPTION (Continued)
When ordering parts, state Model, Part Number, and Description

For SAFETY use only equivalent replacement part.
(1) Used in 13" Models. RAX150WA01, RAX150WA02, RAX156CH01, RAX156CH02.
(2) Used in 20" Model. RKX185BK04.
(3) Part of 4 Switch Assembly. Part No. 8235 040 50040.
(4) Part of 5 Switch Assembly. Part No. 8234 040 50050.
(5) Part of 5 Switch Assembly. Part No. 8235 040 50060.